We promise that this tutorial is not a permanent and unchanging document. We will continuously revise and optimize the tutorial based on feedback from everyone on the forum and actual development experience.

Order

First of all, thank you all for purchasing the ZYNQ development boards AXU3EG, AXU4EV, and AXU5EV produced by Xinyi Electronic Technology (Shanghai) Co., Ltd.! Your support and trust in us and our products have added confidence and courage to us to move forward forever.

Someone may ask if someone with zero foundation can learn ZYNQ? It depends on where the zero is. If you can't even understand the schematic, don't know what an array is in C language, and have no concept of pointers, then you have a negative foundation. To learn ZYNQ, you need basic hardware knowledge and proficient C language skills.

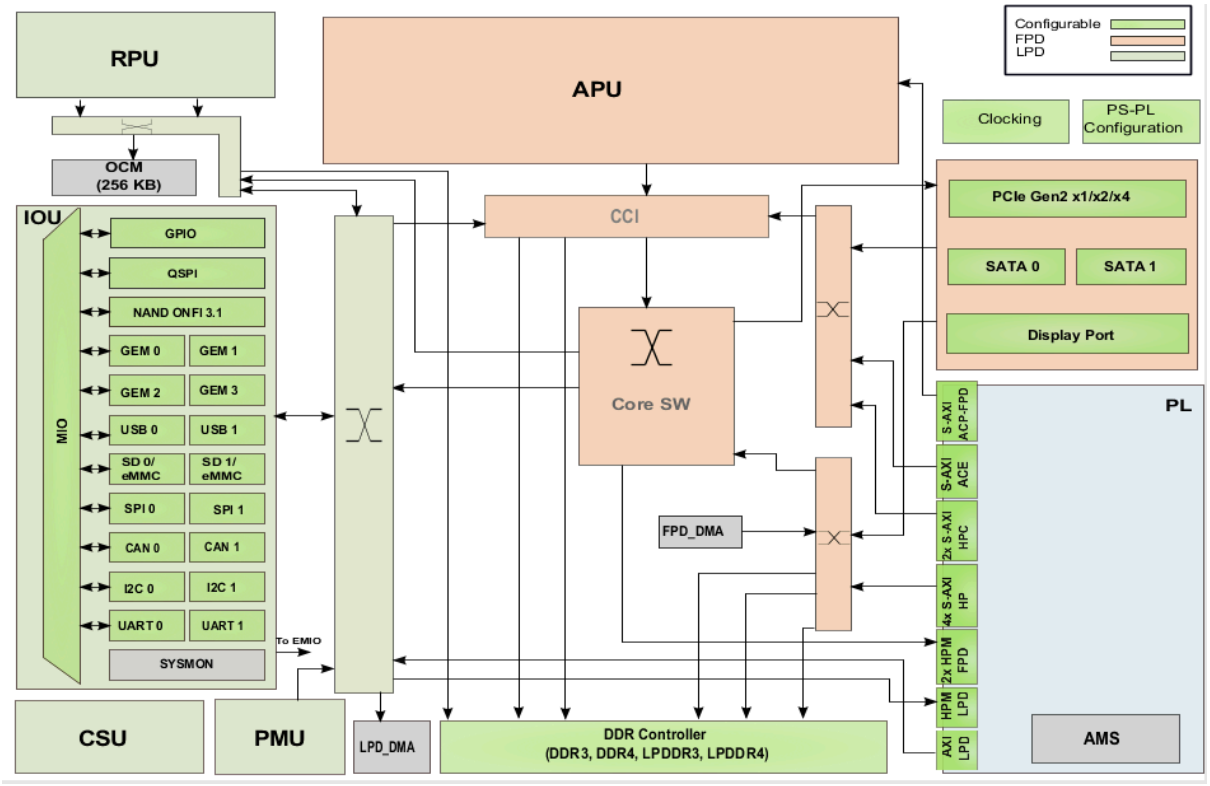
This tutorial is a partial tutorial on FPGA. Through continuous practice, you can master the basic process of FPGA development. Although it does not explain many principles, practice makes perfect. With more practice, you can gradually master the secrets.

Part 1: Introduction to Ultrascale+ MPSoC

The Zynq UltraScale + MPSoC series is the second generation Zynq platform of Xilinx. Its highlight is that the FPGA includes a complete ARM processing subsystem (PS), including a quad-core Cortex-A53 processor or a dual-core Cortex-A53 plus a dual-core Cortex-R5 processor. The entire processor is built around the processor, and the processor subsystem integrates a memory controller and a large number of peripherals, making the processor core completely independent of the programmable logic unit in Zynq. In other words, if the programmable logic unit part (PL) is not currently used, the ARM processor subsystem can also work independently, which is fundamentally different from the previous FPGA, which is processor-centric.

Zynq consists of two major functional blocks, PS and PL. In other words, it is the SOC part of ARM and the FPGA part. Among them, PS integrates APU ARM Cortex ™ - A53 processor, RPU Cortex-R5 processor, AMBA ® interconnect, internal memory (OCM), external memory interface (DDR Controller) and peripherals (IOU). These

peripherals (IOU) mainly include USB bus interface, Ethernet interface, SD/eMMC interface, I2C bus interface, CAN bus interface, UART interface, GPIO, etc. High-speed interfaces such as PCIe, SATA, Display Port.



Overall block diagram of ZYNQ MPSoC chip

PS: Processing System is the part of ARM's SoC that is independent of FPGA.

PL: Programmable Logic, which refers to the FPGA part.

1.

PS and PL interconnection technology

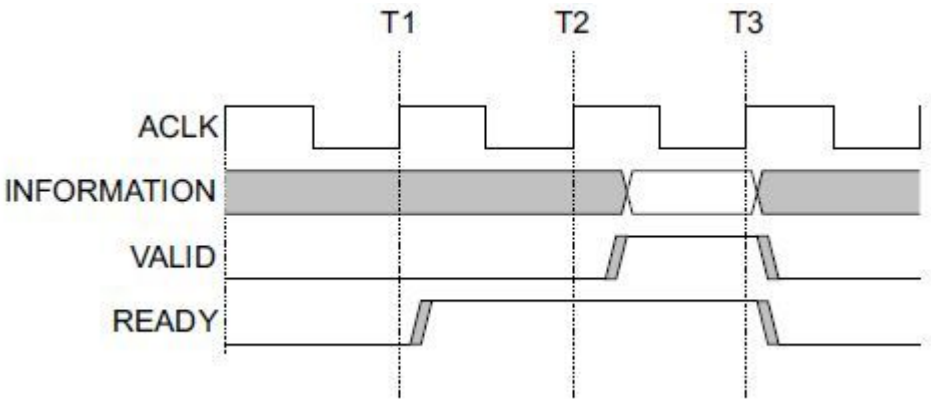
ZYNQ, as a product that closely combines high-performance ARM Cortex-A53 series processors with high-performance FPGA in a single chip, needs to design an efficient interconnection path between on-chip high-performance processors and FPGA to achieve high-speed communication and data exchange between ARM processors and FPGA, and to leverage the performance advantages of ARM processors and FPGA. Therefore, how

to design an efficient PL and PS data interaction path is the top priority of ZYNQ chip design and one of the keys to the success or failure of product design. In this section, we will mainly introduce the connection between PS and PL, allowing users to understand the technology of the connection between PS and PL.

In fact, in the specific design, we often don't need to do too much work on the connection. After we add the IP core, the system will automatically use the AXI interface to connect our IP core to the processor. We just need to add a little more.

AXI, short for Advanced eXtensible Interface, is an interface protocol introduced by Xilinx starting from the FPGA series 6. It mainly describes the data transmission method between master and slave devices. It continues to be used in ZYNQ, with version AXI4, so we often see AXI4.0. ZYNQ internal devices all have AXI interfaces. In fact, AXI is a part of AMBA (Advanced Microcontroller Bus Architecture) proposed by ARM, which is a high-performance, high-bandwidth, low-latency on-chip bus and is also used to replace the previous AHB and APB buses. The first version of AXI (AXI3) was included in AMBA3.0 released in 2003, and the second version of AXI (AXI4) was included in AMBA 4.0 released in 2010.

The AXI protocol mainly describes the data transmission method between the master and slave devices. The master and slave devices establish a connection through handshake signals. When the slave device is ready to receive data, it will send a READY signal. When the data of the master device is ready, it will send and maintain a VALID signal, indicating that the data is valid. Data only starts to be transmitted when both VALID and READY signals are valid. When these two signals remain valid, the master device will continue to transmit the next data. The master device can revoke the VALID signal or the slave device can revoke the READY signal to terminate the transmission. The AXI protocol is shown in the figure. At T2, the READY signal of the slave device is valid, and at T3, the VILID signal of the master device is valid, and the data transmission begins.



It has the characteristics of lightweight and simple structure, suitable for small batch data and simple control occasions. It does not support batch transmission, and can only read and write one word (32 bits) at a time. It is mainly used to access some low-speed peripherals and control peripherals.

AXI4:

The interface is similar to AXI-Lite, except that it adds a function of batch transmission, which can continuously read and write one address at a time. That is to say, it has the burst function of data reading and writing.

Both of the above methods use memory mapping control, that is, ARM weaves the user-defined IP into a certain address for access. When reading and writing, it is like reading and writing to your own on-chip RAM, and programming is also very convenient, with low development difficulty. The cost is that it occupies too many resources and requires additional signal lines such as read address lines, write address lines, read data lines, write data lines, and write response lines.

AXI4-Stream：

This is a continuous stream interface that does not require address lines (similar to FIFO, just keep reading or writing). For this type of IP, ARM cannot control it through the memory mapping method above (FIFO has no concept of address), and there must be a conversion device, such as AXI-DMA module, to achieve the conversion from memory mapping to streaming interface. AXI-Stream is suitable for many occasions: video stream processing, Communication Protocol conversion, Digital Signal Processing, wireless communication, etc. Its essence is a data path constructed for numerical streams, constructing a continuous data stream from the source (such as ARM memory, DMA, wireless receiving front end, etc.) to the sink (such as HDMI display, high-speed AD audio output, etc.). This interface is suitable for real-time signal processing.

AXI4 and AXI4-Lite interfaces contain 5 different channels.

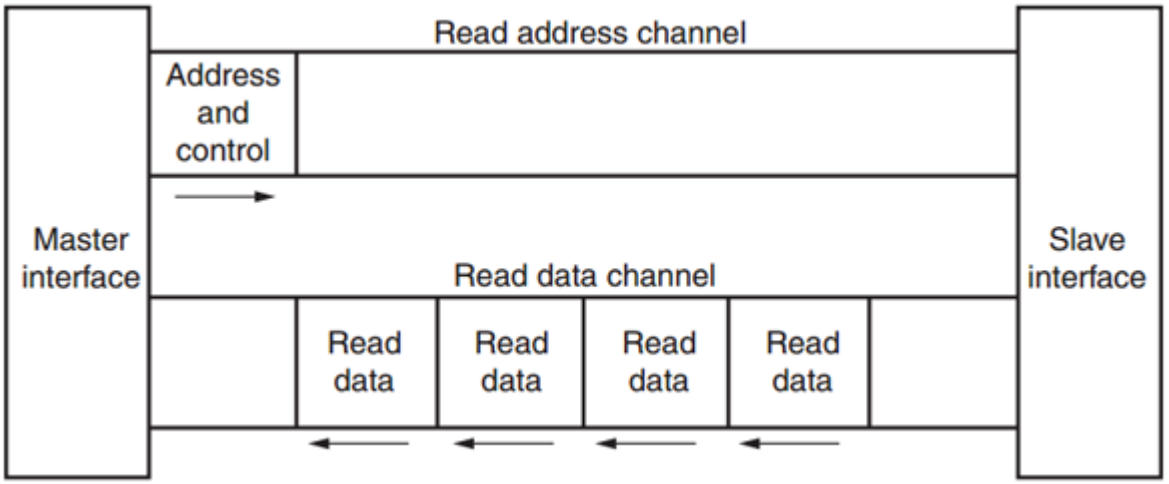
- Read Address Channel
- Write Address Channel
- Read Data Channel
-

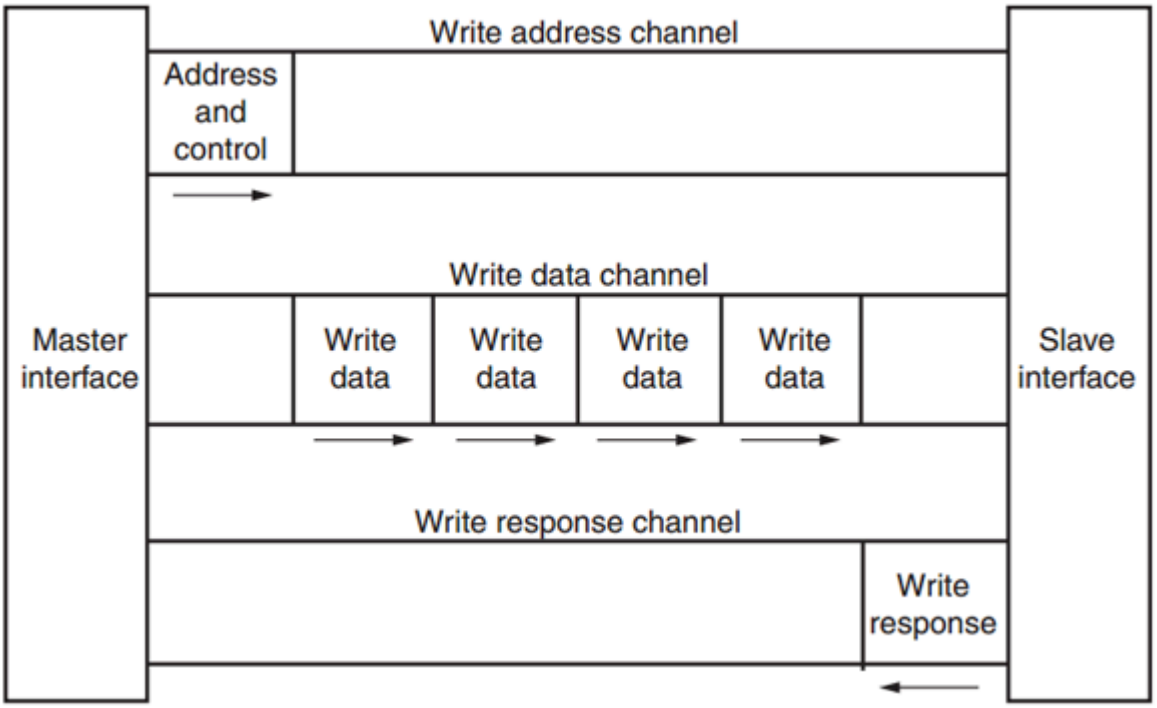
Write Data Channel

-

Write Response Channel

Each channel is an independent AXI handshake protocol. The following two figures show the read and write models respectively.





AXI write data channel

The AXI bus protocol is implemented by hardware in ZYNQ chip, including 12 physical interfaces, namely S_AXI_HP {0:3} _FPD, S_AXI_LPD, S_AXI_ACE_FPD, S_AXI_ACP_FPD, S_AXI_HPC {0,1} _FPD, M_AXI_HPM {0,1} _FPD, M_AXI_HPM0_LPD interfaces.

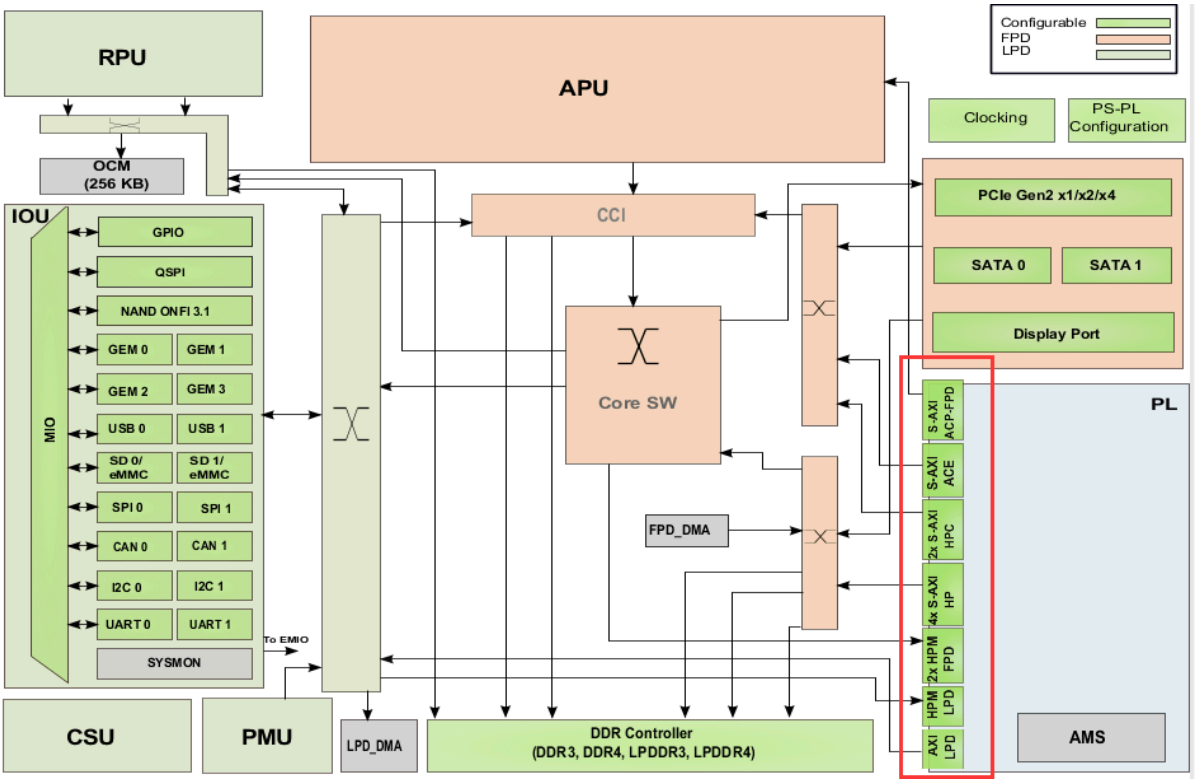
S_AXI_HP {0:3} _FPD interface, is a high-performance/bandwidth AXI4 standard interface, a total of four, PL module as the main device connection. Mainly used for PL access to memory on PS (DDR and FPD Main Switch)

S_AXI_LPD interface, high-performance port, connecting PL to LPD. Low-latency access to OCM and TCM, access to PS-side DDR.

S_AXI_HPC {0,1} _FPD interface, connect PL to FPD, can connect to CCI, access L1 and L2 Cache, due to CCI, access to DDR controller will have a large delay.

M_AXI_HPM {0,1} _FPD interface, high-performance bus, PS is master, connect FPD to PL, can be used for CPU, DMA, PCIe, etc. to push a large amount of data from PS to PL.

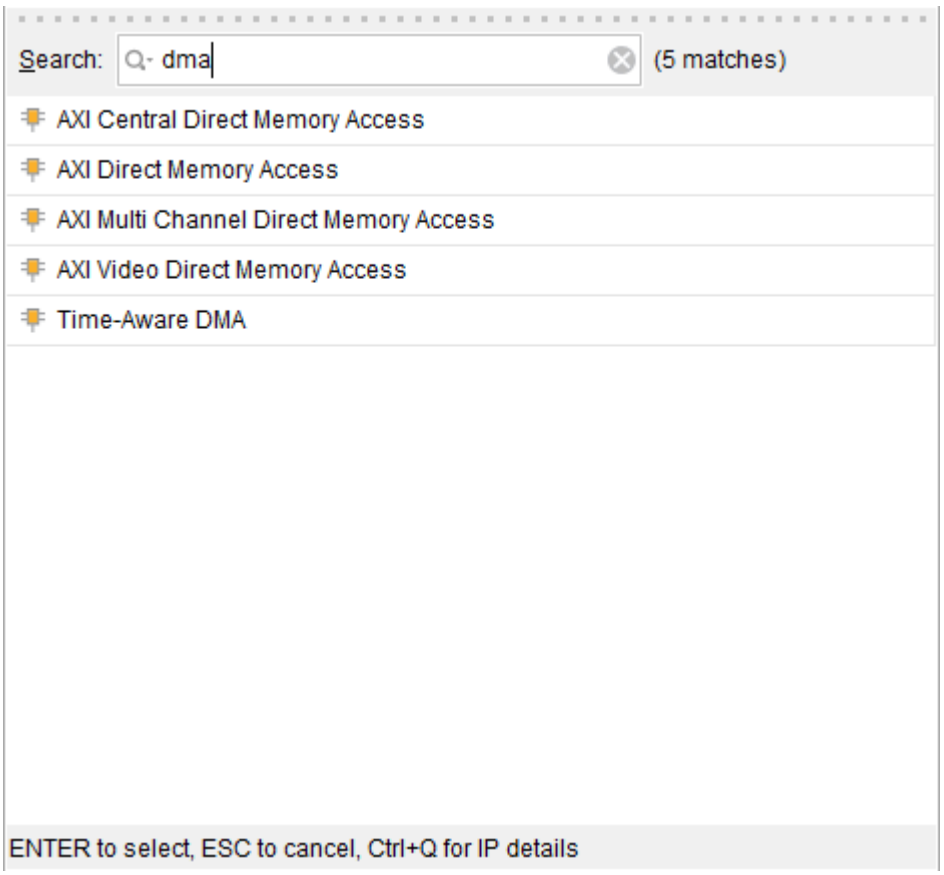
M_AXI_HPM0_LPD interface, low latency interface bus, PS is master, connecting LPD to PL, can directly access the PL end of the BRAM, DDR, etc., is also often used to configure the PL end of the register.



Only M_AXI_HPM {0,1} _FPD and M_AXI_HPM0_LPD are Master Ports, that is, the host interface, and the rest are Slave Ports (slave interfaces). The host interface has the permission to initiate read and write, and ARM can use the two host interfaces to

actively access PL logic. In fact, it maps PL to a certain address, and reading and writing PL registers is like reading and writing its own memory. The rest of the slave interfaces belong to passive interfaces, which accept read and write from PL and accept it. In PS and PL interconnection applications, the most commonly used interfaces are S_AXI_HP {0:3} _FPD, M_AXI_HPM {0,1} _FPD and M_AXI_HPM0_LPD.

ARM located on the PS side directly supports the AXI interface with hardware, while PL needs to use logic to implement the corresponding AXI protocol. Xilinx provides ready-made IPs such as AXI-DMA, AXI-GPIO, AXI-Dataover, and AXI-Stream in the Vivado Development Environment, all of which implement corresponding interfaces. When using them, you can directly add them from the IP list of Vivado to achieve the corresponding functions. The following picture shows various DMA IPs under Vivado.



Below is an introduction to the functions of several commonly used AXI interface IPs.

AXI-DMA: Conversion from PS memory to PL high-speed transmission high-speed channel AXI-HP < ---- > AXI-Stream

AXI-FIFO-MM2S: Conversion from PS memory to PL universal transport channel AXI-HPM <-----> AXI-Stream

AXI-Datamover: realizes the conversion from PS memory to PL high-speed transmission high-speed channel AXI-HP < ---- > AXI-Stream, but this time it is completely controlled by PL, and PS is completely passive.

AXI-VDMA: realizes the conversion from PS memory to PL high-speed transmission high-speed channel AXI-HP < ---- > AXI-Stream, but it is specifically designed for two-dimensional data such as videos and images.

AXI-CDMA: This is done by PL to move data from one location in memory to another without CPU intervention.

We will give examples of how to use these IPs in later chapters. Sometimes, users need to develop their own defined IPs to communicate with PS, and they can use the wizard to generate corresponding IPs. User-defined IP cores can have interfaces such as AXI4-Lite, AXI4, AXI-Stream, PLB, and FSL. The latter two are not used because ARM does not support them.

With the above official IPs and custom IPs generated by the wizard, users don't actually need to know too much about AXI timing (unless they really encounter problems), because Xilinx has encapsulated all the details related to AXI timing, and users only need to focus on their own logic implementation.

Strictly speaking, the AXI protocol is a peer-to-peer leader/follower interface protocol. When multiple peripherals need to interact with each other, we need to add an AXI Interconnect module, which is the AXI interconnection matrix. Its function is to provide a switching mechanism for connecting one or more AXI master devices to one or more AXI slave devices (similar to the switching matrix in switch).

This AXI Interconnect IP core can support up to 16 master devices and 16 slave devices. If more interfaces are needed, multiple IP cores can be added.

There are several basic connection modes for AXI Interconnect:

- N-to-1 Interconnect
- to-N Interconnect
- N-to-M Interconnect (Crossbar Mode)
- N-to-M Interconnect (Shared Access Mode)

The AXI interface devices inside ZYNQ are interconnected through the interconnection matrix, which ensures the efficiency of data transmission and the flexibility of connection. In Vivado, Xilinx provides an IP core axi_interconnect to implement this interconnection matrix, which we only need to call.

- Tcl script
-

Good foundation in English reading

3.2

Logic developers

-

Computer organization principle

-

C language

-

Fundamentals of digital circuits

Part 2: Development board hardware introduction

Xinyi Electronic Technology (Shanghai) Co., Ltd. has officially released the 2019 development board (model: AXU4EV-P) based on the XILINX Zynq UltraScale + MPSoCs development platform. In order to help you quickly understand this development platform, we have compiled this user manual.

This MPSoCs development platform adopts a core board and expansion board mode, which facilitates users' secondary development and utilization of the core board. The core board uses the XILINX Zynq UltraScale + EV chip ZU4EV solution, which

integrates quad-core ARM Cortex-A53 and FPGA programmable logic on one chip using Processing System (PS) + Programmable Logic (PL) technology. In addition, the PS end of the core board has four 4GB high-speed DDR4 SDRAM chips, one 8GB eMMC storage chip, and one 256Mb QSPI FLASH chip; the PL end of the core board has one 1GB DDR4 SDRAM chip.

We have expanded a rich range of peripheral interfaces for users in baseplate design, such as 1 FMC LPC interface, 1 SATA M.2 interface, 1 DP interface, 1 PCIE x2 interface, 2 SFP + fiber optic interfaces, 4 USB3.0 interfaces, 2 Gigabit Ethernet interfaces, 1 HDMI input interface, 1 HDMI output interface, 2 UART interfaces, 1 SD card interface, 2 CAN bus interfaces, 2 RS485 interfaces, and 1 MIPI camera interface, etc. It meets users' requirements for various high-speed data exchange, data storage, video transmission processing, Deep learning, artificial intelligence, and industrial control, and is a "professional-level" ZYNQ development platform. It provides the possibility for high-speed data transmission and exchange, early verification and later application of data processing. It is believed that such a product is very suitable for students, engineers and other groups engaged in MPSoCs development.

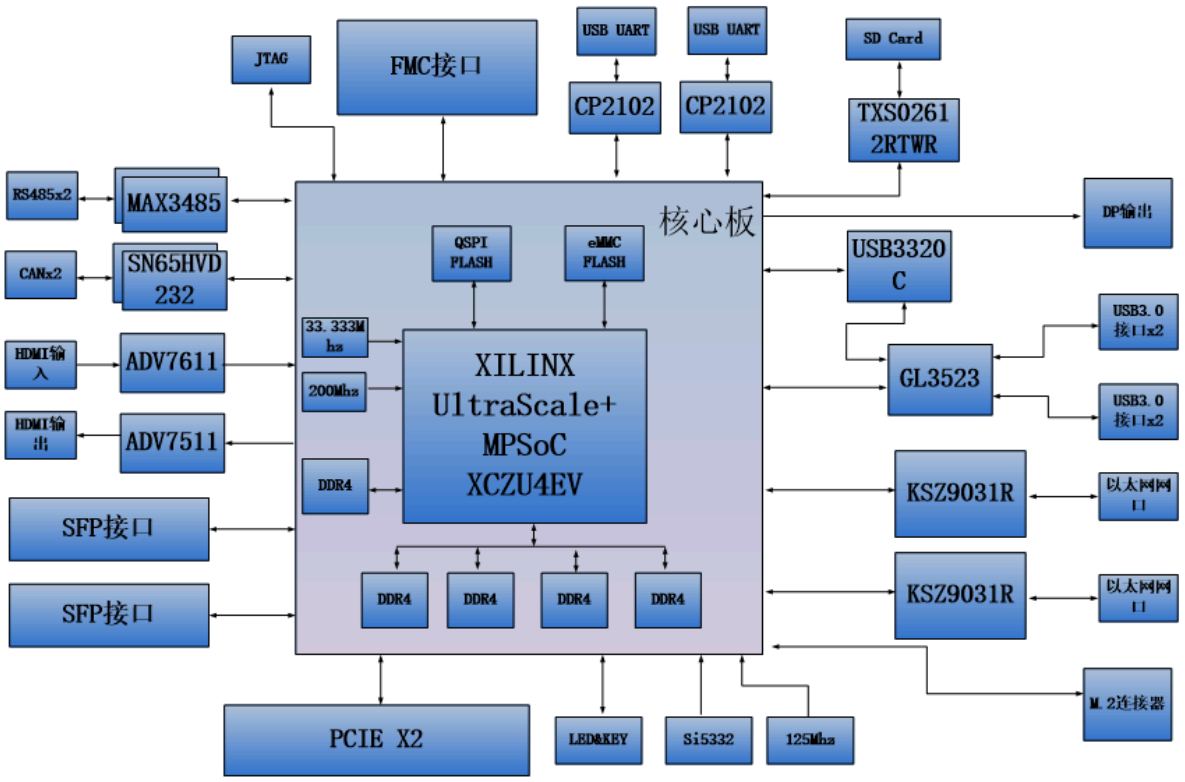
Here is a brief introduction to the functions of the AXU4EV-P MPSoCs development platform.

The entire structure of the development board inherits our consistent core board + expansion board pattern. The core board and expansion board are connected by high-speed board-to-board connectors.

The core board is mainly composed of the smallest system of ZU4EV + 5 DDR4 + eMMC + 1 QSPI FLASH. ZU4EV uses Xilinx's Zynq UltraScale + MPSoCs EV series chip, model XCZU4EV-1SFVC784I. The ZU4EV chip can be divided into the processor system part Processor System (PS) and the programmable logic part Programmable Logic (PL). Four and one DDR4 chips are respectively hung on the PS and PL ends of the ZU4EV chip, with each DDR4 chip having a capacity of up to 1G bytes, enabling the ARM system and FPGA system to independently process and store data. The 8GB eMMC FLASH memory chip and one 256Mb QSPI FLASH chip on the PS end are used to statically store the operating system, file system, and user data of MPSoCs.

The baseplate has expanded the core board with rich peripheral interfaces, including 1 M.2 interface, 1 DP output interface, 4 USB3.0 interfaces, 2 Gigabit Ethernet interfaces, 1 PCIE x2 interface, 2 fiber optic interfaces, 2 UART interfaces, 1 SD card interface, 1 HDMI output interface, 1 HDMI input interface, 1 FMC interface, 2 CAN bus interfaces, 2 RS485 interfaces, 1 MIPI interface, and some button LEDs.

The following is a schematic diagram of the structure of the entire development system.



-

JTAG debugging port

One 10-pin 2.54mm standard JTAG port is used for downloading and debugging FPGA programs. Users can debug and download the ZU4EV system through the XILINX downloader.

-

Temperature and humidity sensor

One temperature and humidity sensor chip LM75 is onboard to detect the temperature and humidity of the surrounding environment of the board.

-

EEPROM

EEPROM 24LC04 with 1 IIC interface.

-

RTC real-time clock

1 built-in RTC real-time clock;

-

LED lights

5 Light Emitting Diode LEDs, 2 on the core board and 3 on the baseplate. 1 power indicator and 1 DONE configuration indicator on the core board. 1 power indicator and 2 user indicator on the baseplate.

-

Button

3 buttons, 1 reset button, 2 user buttons.

1.

ACU4EV Core Board

1.1

Introduction

ACU4EV (**core board model, the same below**) core board, ZYNQ chip is based on XILINX's Zynq UltraScale + MPSoCs EV series XCZU4EV-1SFVC784I.

This core board uses 5 Micron DDR4 chips MT40A512M16GE, with 4 DDR4 chips mounted on the PS side, forming a 64-bit data bus bandwidth and a capacity of 4GB. 1 chip is mounted on the PL side, with a 16-bit data bus width and a capacity of 1GB. The maximum operating speed of the DDR4 SDRAM on the PS side can reach 1200MHz (data rate 2400Mbps), and the maximum operating speed of the DDR4 SDRAM on the PL side can reach 1066MHz (data rate 2132Mbps). In addition, the core board also integrates a 256MBit QSPI FLASH and an 8GB eMMC FLASH chip for starting storage configuration and system files.

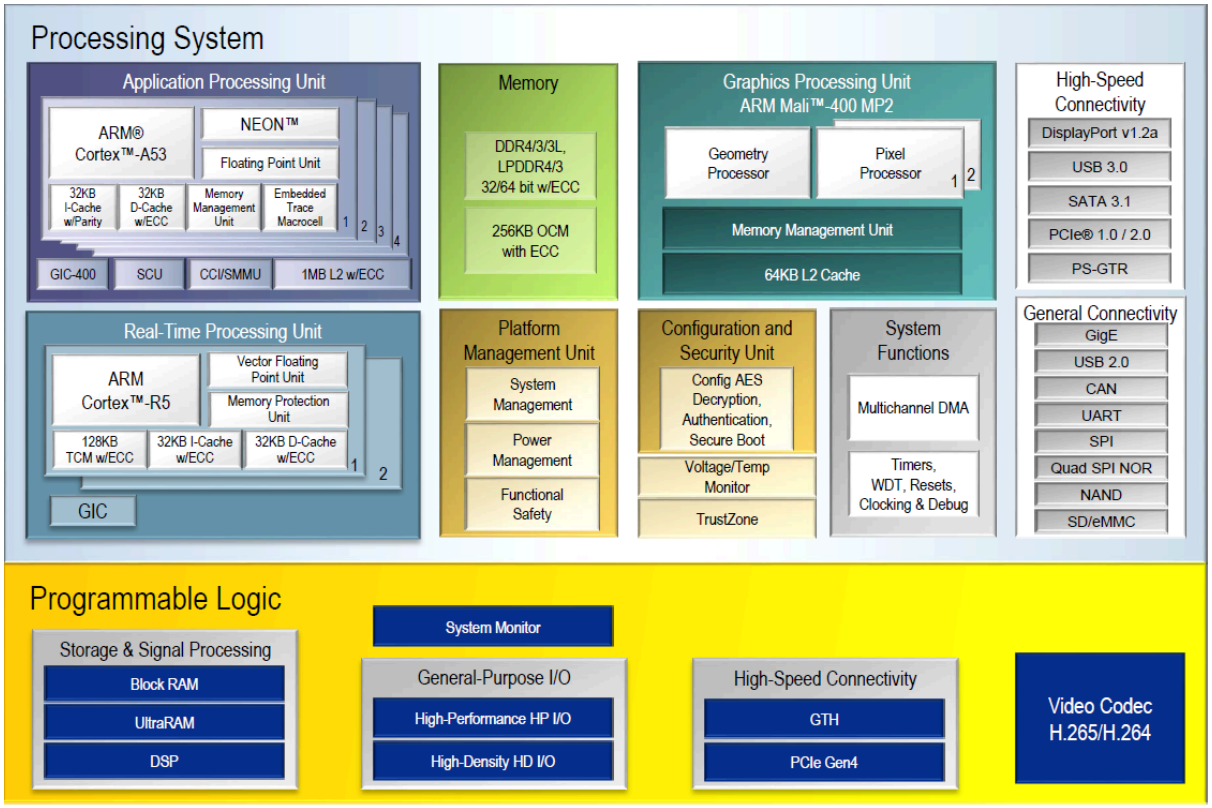
In order to connect with the baseplate, the four board-to-board connectors of this core board have expanded the USB 2.0 interface, Gigabit Ethernet interface, SD card interface, and other remaining MIO ports on the PS side; it has also expanded four pairs of PS MGT high-speed transceiver interfaces; and the four GTH transceivers and almost all IO ports (HP I/O: 96, HD I/O: 84) on the PL side. The wiring between the XCZU4EV chip and the interface has been processed equally and differentially, and the core board size is only 80 * 60 (mm), which is very suitable for secondary development.

1.2

ZYNQ chip

The development board uses Xilinx's Zynq UltraScale + MPSoCs EV series chip, model XCZU4EV-1SFVC784I. The PS system of the ZU4EV chip integrates four ARM Cortex ™ - A53 processors, with a speed of up to 1.5Ghz and supports level 2 cache; in addition, ZU4EV also includes two Cortex-R5 processors (with a speed of up to 600Mhz) and one video codec processor.

The ZU4EV chip supports 32-bit or 64-bit DDR4, LPDDR4, DDR3, DDR3L, LPDDR3 storage chips, with rich high-speed interfaces such as PCIE Gen2, USB3.0, SATA 3.1, DisplayPort on the PS side; it also supports USB2.0, Gigabit Ethernet, SD/SDIO, I2C, CAN, UART, GPIO and other interfaces. The PL side contains rich programmable logic units, DSP and internal RAM. The overall block diagram of the ZU4EV chip is shown in Figure 2-2-1



-

DSP Slices: 728

-

Image Codec Unit (VCU): 1

-

PCIE3.0:2 pieces

-

GTH 12.5Gb/s transceiver: 4

The speed level of the XCZU4EV-1SFVC784I chip is -1, industrial grade, and packaged as SFVC784.

1.3

DDR4 DRAM

The ACU4EV core board is equipped with 5 Micron (Micron) 1GB DDR4 chips, model MT40A512M16GE-083E, among which 4 DDR4 chips are mounted on the PS side, forming a 64-bit data bus bandwidth and 4GB capacity. 1 chip is mounted on the PL side, with a 16-bit data bus width and 1GB capacity. The maximum operating speed of the DDR4 SDRAM on the PS side can reach 1200MHz (data rate 2400Mbps), and the 4 DDR4 storage systems are directly connected to the memory interface of PS's BANK504. The maximum operating speed of the DDR4 SDRAM on the PL side can reach 1066MHz (data rate 2132Mbps), and 1 DDR4 chip is connected to the interface of FPGA's BANK64. The specific configuration of the DDR4 SDRAM is shown in the following table 2-3-1.

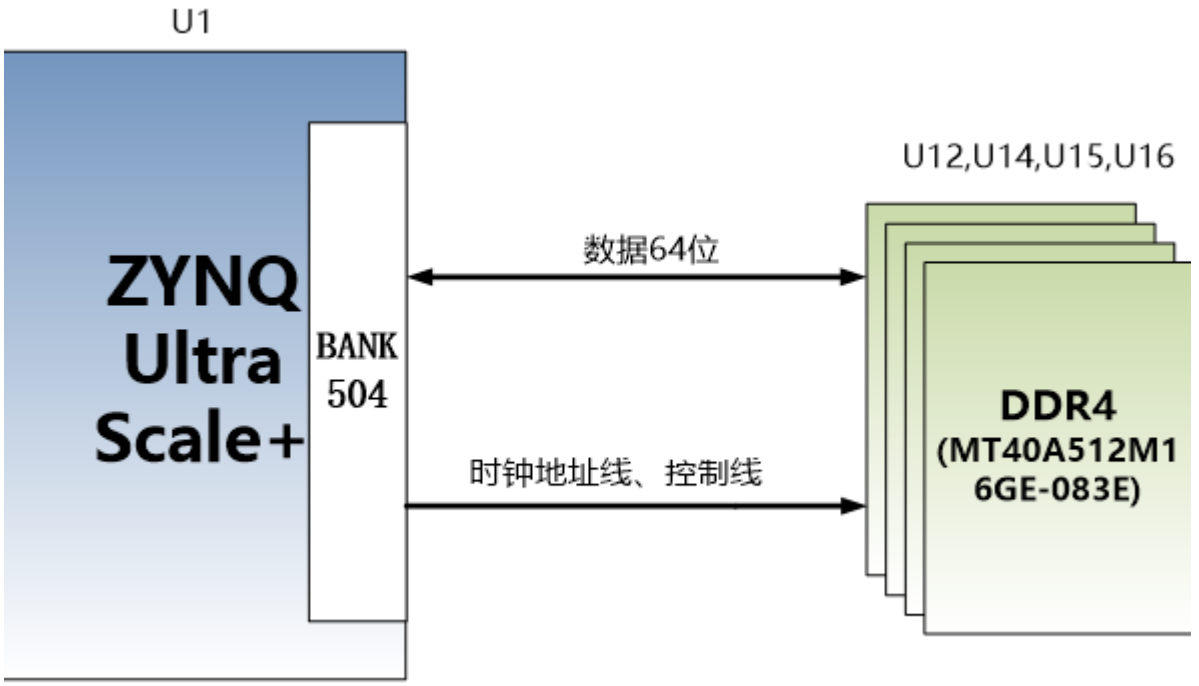
Locus number	Chip model	Capacity	Manufacturer

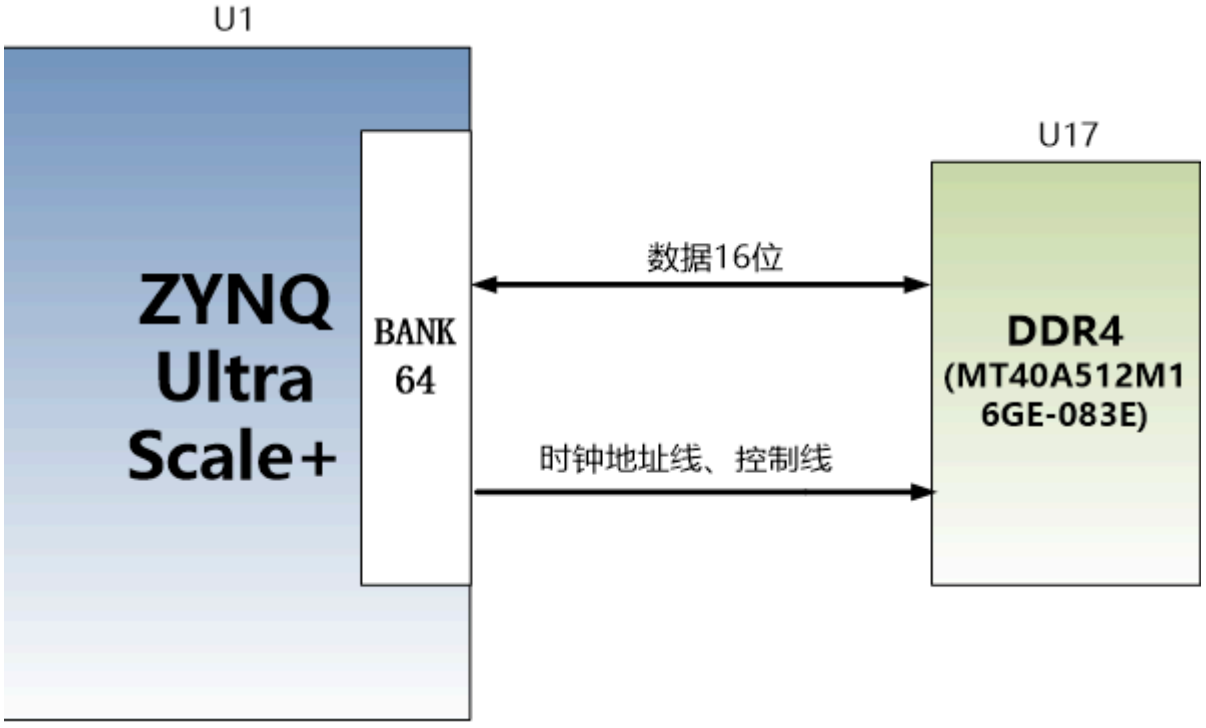
U12,U14,U15,U16	IT40A512M16GE-083E	512M x 16bit	Micron

Table 2-3-1 DDR4 SDRAM Configuration

The hardware design of DDR4 requires strict consideration of signal integrity. We have fully considered matching resistors/end point resistors, wiring impedance control, and wiring length control in circuit design and PCB design to ensure the high-speed and stable operation of DDR4.

PS DDR4 hardware connection is shown in Figure 2-3-1:


U1
ZYNQ Ultra Scale+
BANK 504
数据64位
时钟地址线、控制线
U12,U14,U15,U16
DDR4 (MT40A512M16GE-083E)


U1
ZYNQ Ultra Scale+
BANK 64
数据16位
时钟地址线、控制线
U17
DDR4 (MT40A512M16GE-083E)

PL_DDR4_A3	IO_L3N_T0L_N5_AD15N_64	AC8
PL_DDR4_A4	IO_L11P_T1U_N8_GC_64	AF7
PL_DDR4_A5	O_L4P_T0U_N6_DBC_AD7P_64	AD7

PL_DDR4_A6	IO_L9N_T1L_N5_AD12N_64	AH7
PL_DDR4_A7	IO_L2P_T0L_N2_64	AE9
PL_DDR4_A8	IO_L9P_T1L_N4_AD12P_64	AH8
PL_DDR4_A9	IO_L1P_T0L_N0_DBC_64	AC9
PL_DDR4_A10	O_L4N_T0U_N7_DBC_AD7N_64	AE7
PL_DDR4_A11	O_L7N_T1L_N1_QBC_AD13N_64	AH9

PL_DDR4_A12	IO_L6N_T0U_N11_AD6N_64	AC6
PL_DDR4_A13	IO_L1N_T0L_N1_DBC_64	AD9
PL_DDR4_BA0	IO_T1U_N12_64	AH6
PL_DDR4_BA1	IO_L5N_T0U_N9_AD14N_64	AC7
PL_DDR4_RAS_B	IO_T2U_N12_64	AB5
PL_DDR4_CAS_B	IO_L5P_T0U_N8_AD14P_64	AB7
PL_DDR4_WE_B	IO_L11N_T1U_N9_GC_64	AF6

PL_DDR4_ACT_B	O_L13N_T2L_N1_GC_QBC_64	AD4
PL_DDR4_CS_B	IO_L6P_T0U_N10_AD6P_64	AB6
PL_DDR4_BG0	IO_L2N_T0L_N3_64	AE8
PL_DDR4_RST	O_L7P_T1L_N0_QBC_AD13P_64	AG9
PL_DDR4_CLK_N	O_L10N_T1U_N7_QBC_AD4N_64	AG5
PL_DDR4_CLK_P	O_L10P_T1U_N6_QBC_AD4P_64	AG6

PL_DDR4_CKE	IO_T3U_N12_64	AE4
PL_DDR4_OTD	O_L19N_T3L_N1_DBC_AD9N_64	AH4

1.4

QSPI Flash

The ACU4EV core board is equipped with a 256MBit Quad-SPI FLASH chip to form an 8-bit bandwidth data bus. The FLASH model is MT25QU256ABA1EW9, which uses the 1.8V CMOS voltage standard. Due to the non-volatile nature of QSPI FLASH, it can be used as a system startup device to store system startup mirroring. These mirroring mainly include FPGA bit files, ARM application code, and other user data files. The specific model and related parameters of QSPI FLASH are shown in Table 2-4-1.

Locus number	Chip type	Capacity	Manufacturer
U5	MT25QU256ABA1EW9	256M bit	Winbond

Table 2-4-1 QSPI Flash models and parameters

QSPI FLASH is connected to the GPIO port of the PS part of the ZYNQ chip BANK500. In the system design, it is necessary to configure the GPIO port function of these PS ends as QSPI FLASH interface. Figure 4-1 shows the part of QSPI Flash in the schematic diagram.

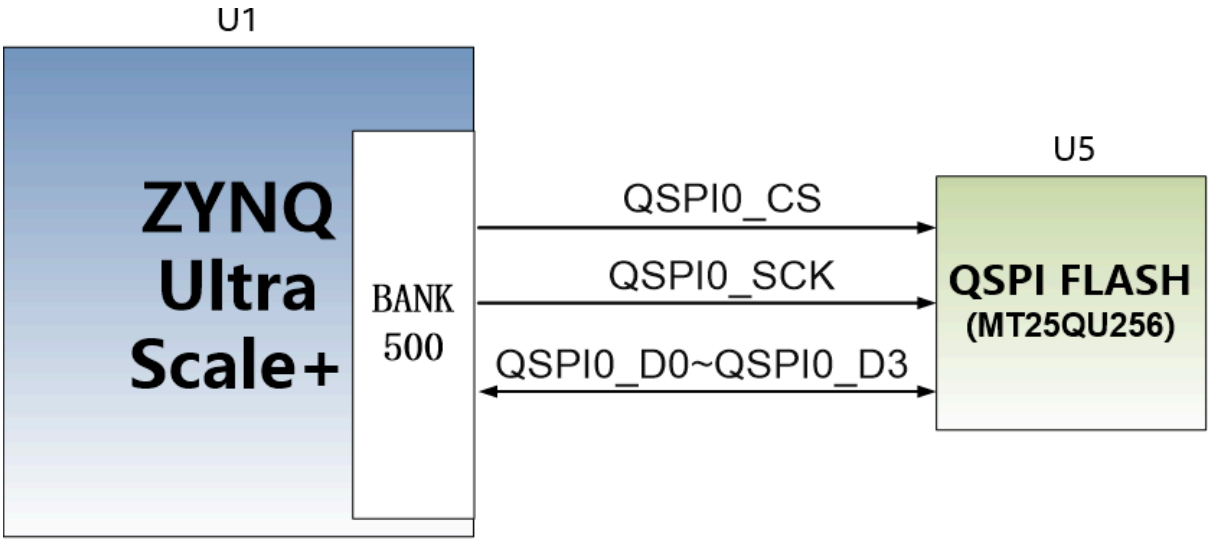


Figure 2-4-1 QSPI Flash connection diagram

Configure chip pin allocation:

Signal name	Pin name	Pin number

MIO0_QSPI0_SCORE	PS_MIO0_500	AG15
MIO1_QSPI0_IO1	PS_MIO1_500	AG16
MIO2_QSPI0_IO2	PS_MIO2_500	AF15
MIO3_QSPI0_IO3	PS_MIO3_500	AH15

MY4_QSPI0_IO0	PS_MIO4_500	AH16
MIO5_QSPI0_SS_B	PS_MIO5_500	AD16

1.5

eMMC Flash

The ACU4EV core board is equipped with a large-capacity 8GB eMMC FLASH chip, model MTFC8GAKAJCN-4M, which supports the HS-MMC interface of JEDEC e-MMC V5.0 standard, with a level support of 1.8V or 3.3V. The data width of eMMC FLASH and ZYNQ connection is 8 bits. Due to the large capacity and non-volatile characteristics of eMMC FLASH, it can be used as a large-capacity storage device in ZYNQ system, such as storing ARM applications, system files, and other user data files. the specific model and related parameters of eMMC FLASH are shown in Table 2-5-1.

Locus number	Chip type	Capacity	Manufacturer
U19	MTFC8GAKAJCN-4M	8G Byte	Micron

Table 2-5-1 eMMC Flash models and parameters

The eMMC FLASH is connected to the GPIO port of the PS part of the ZYNQ UltraScale + BANK500. In the system design, it is necessary to configure the GPIO port function of these PS ends as the EMMC interface. The figure 2-5-1 the part of the eMMC Flash in the schematic diagram.

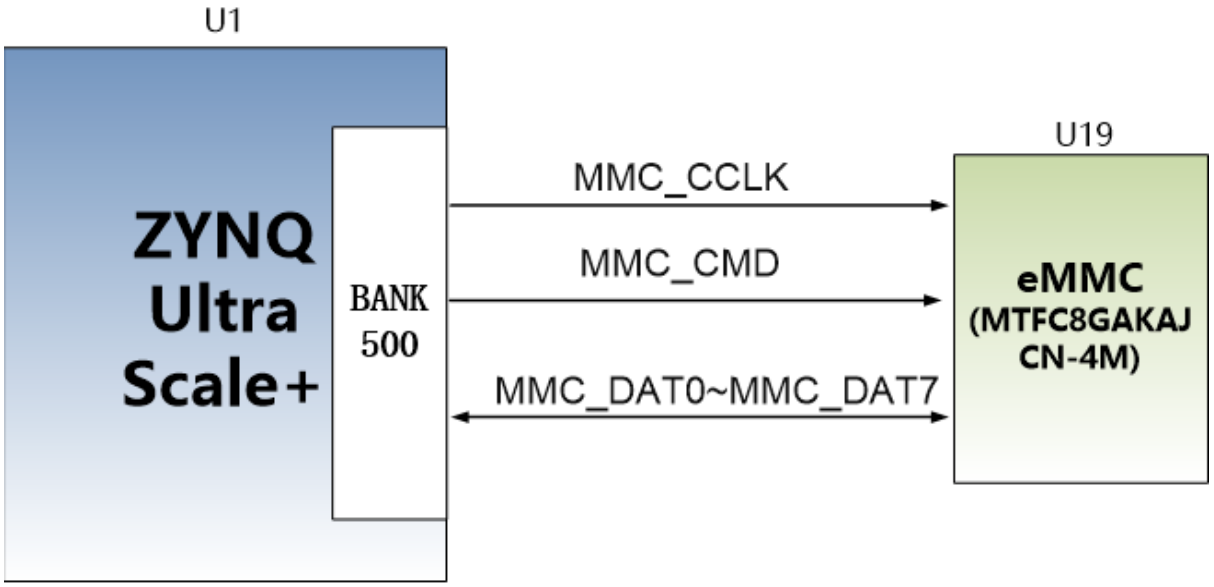


Figure 2-5-1 eMMC Flash connection diagram

Configure chip pin allocation:

Signal name	Pin name	Pin number
MMC_DAT0	PS_MIO13_500	AH18
MMC_DAT1	PS_MIO14_500	AG18

MMC_DAT2	PS_MIO15_500	AE18
MMC_DAT3	PS_MIO16_500	AF18
MMC_DAT4	PS_MIO17_500	AC18
MMC_DAT5	PS_MIO18_500	AC19

MMC_DAT6	PS_MIO19_500	AE19
MMC_DAT7	PS_MIO20_500	AD19
MMC_CMD	PS_MIO21_500	AC21

MMC_CCLK	PS_MIO22_500	AB20
MMC_RSTN	PS_MIO23_500	AB18

1.6

Clock configuration

The core board provides reference clock and RTC real-time clock for PS system and PL logic respectively, so that PS system and PL logic can work independently. The schematic diagram of clock circuit design is shown in the following figure 2-6-1:

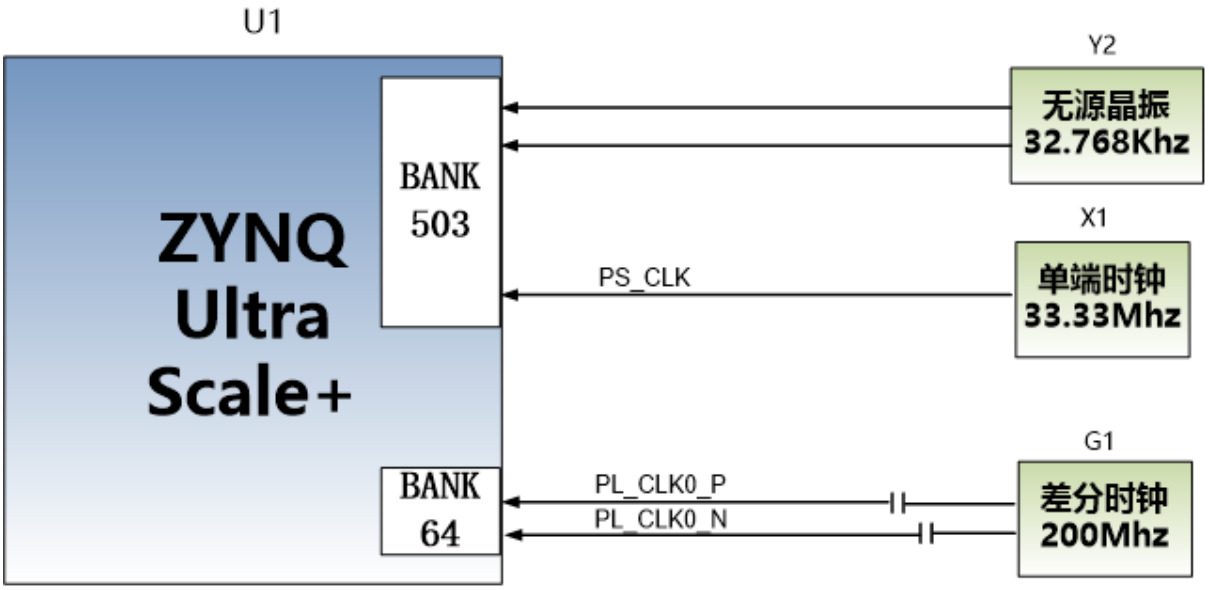
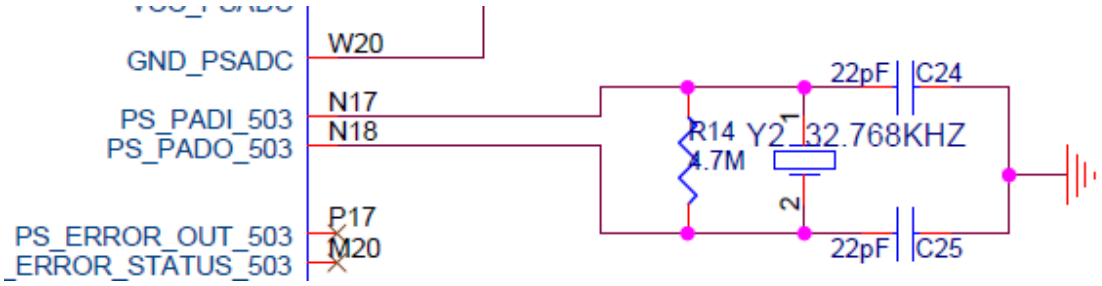


Figure 2-6-1 Core Board Clock Source

PS system RTC real-time clock

The passive crystal Y2 on the core board provides a real-time clock source of 32.768KHz for the PS system. The crystal is connected to the PS_PADI_503 and PS_PADO_503 pins of the BANK503 of the ZYNQ chip. Its schematic is shown in Figure 2-6-2:



Clock pin allocation:

Signal name	Pin
PS_PADI_503	**N17**

PS_PADO_503	**N18**

PS system clock source

The X1 crystal oscillator on the core board provides a clock input of 33.333MHz for the PS part. The clock input is connected to the PS_REF_CLK_503 pin of the BANK503 of the ZYNQ chip. Its schematic is shown in Figure 2-6-3:

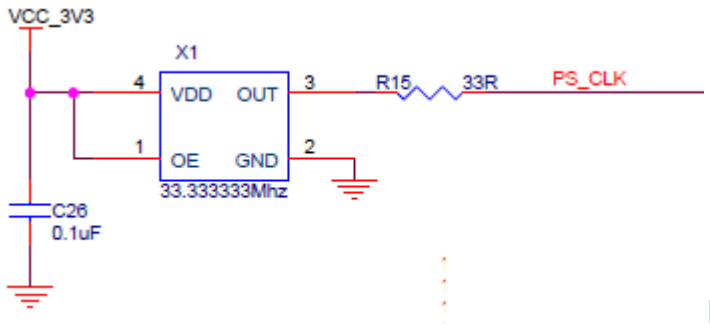


Figure 2-6-3 Active crystal oscillator in PS section

Clock pin allocation:

Signal name	Pin
PS_CLK	R16

PL system clock source

A differential 200MHz PL system clock source is provided on the board for the reference clock of the DDR4 controller. The crystal output is connected to the timestamp oracle (MRCC) of the PL BANK64, which can be used to drive the DDR4 controller and user logic circuit in the FPGA. The schematic of the clock source is shown in Figure 2-6-4

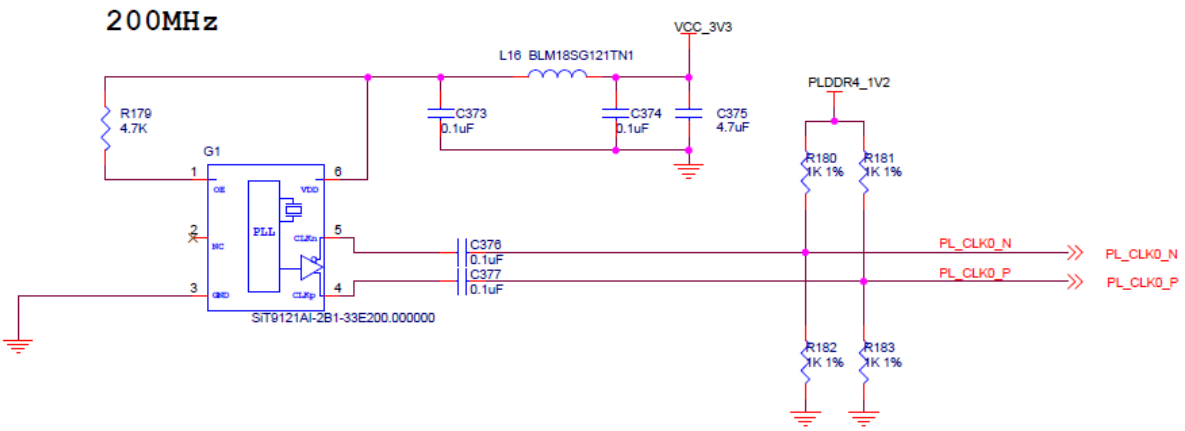


Figure 2-6-4 PL system clock source

PL clock pin allocation:

Signal name	Pin

PL_CLK0_P	AE5
PL_CLK0_N	AF5

1.7

LED lights

The ACU4EV core board has 1 red power indicator (PWR) and 1 configuration LED light (DONE). When the core board is powered, the power indicator light will light up; when the FPGA configuration program, the configuration LED light will light up. The

schematic diagram of the hardware connection of the LED light is shown in Figure 2-7-1:

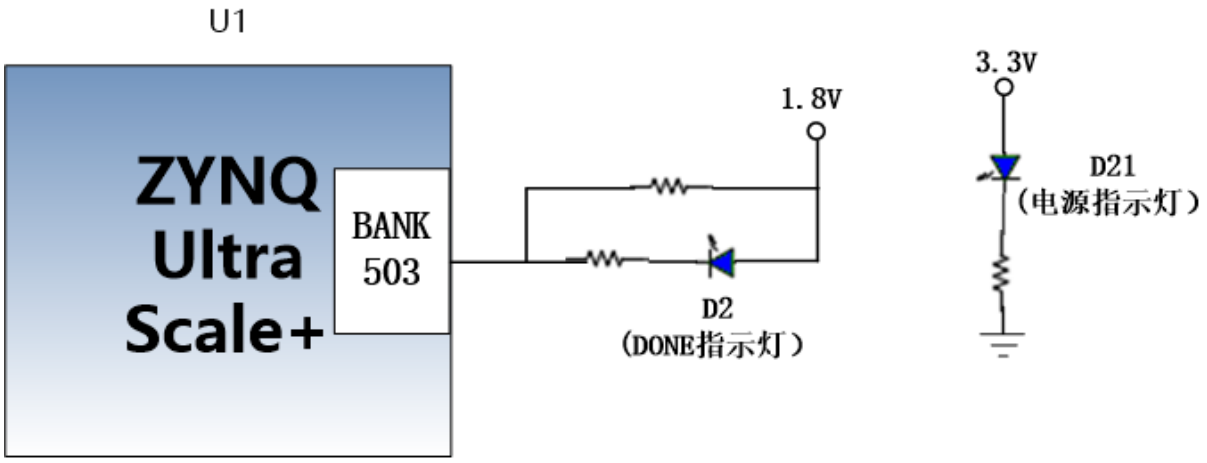
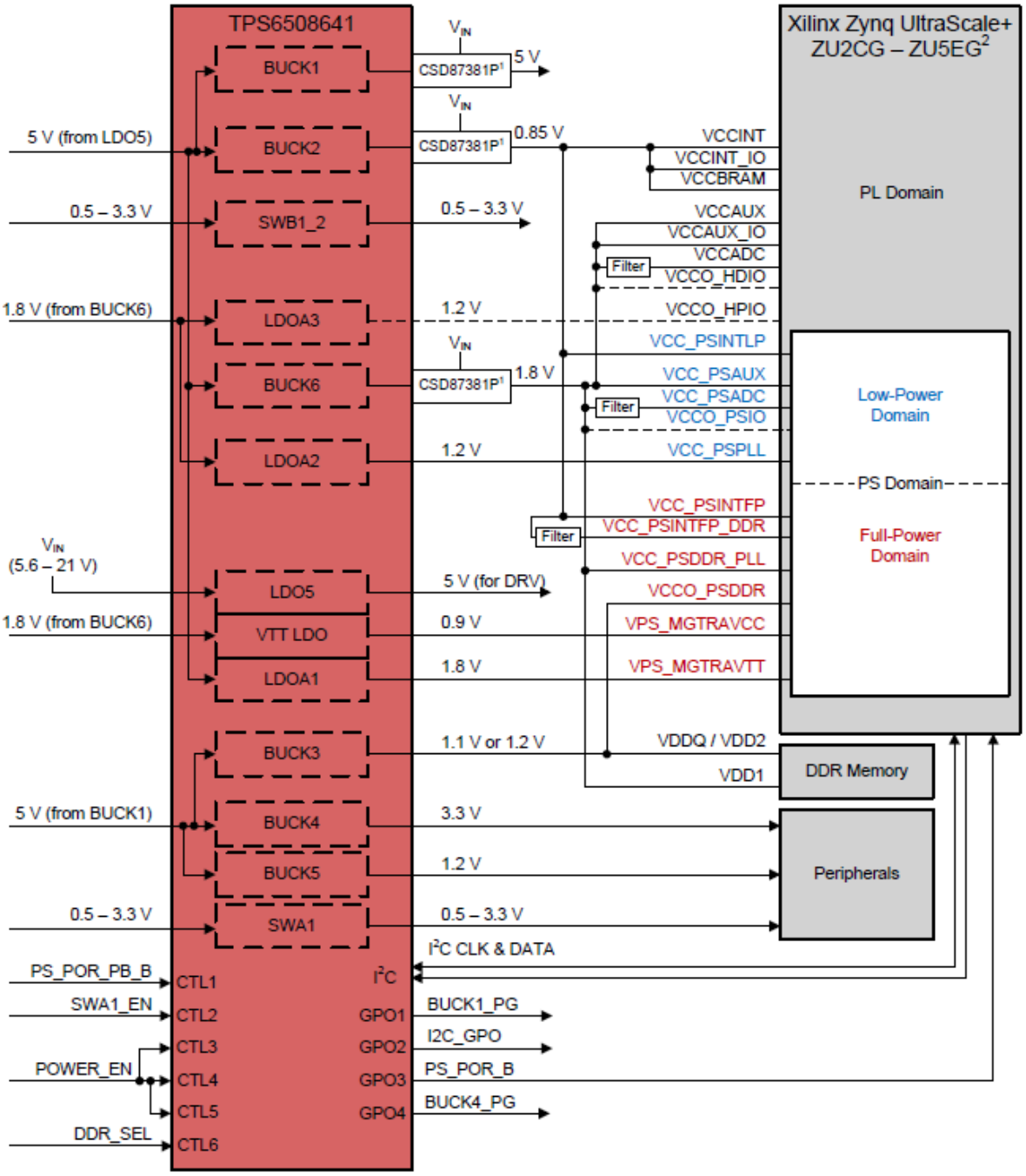


Figure 2-7-1 Core board LED lamp hardware connection schematic

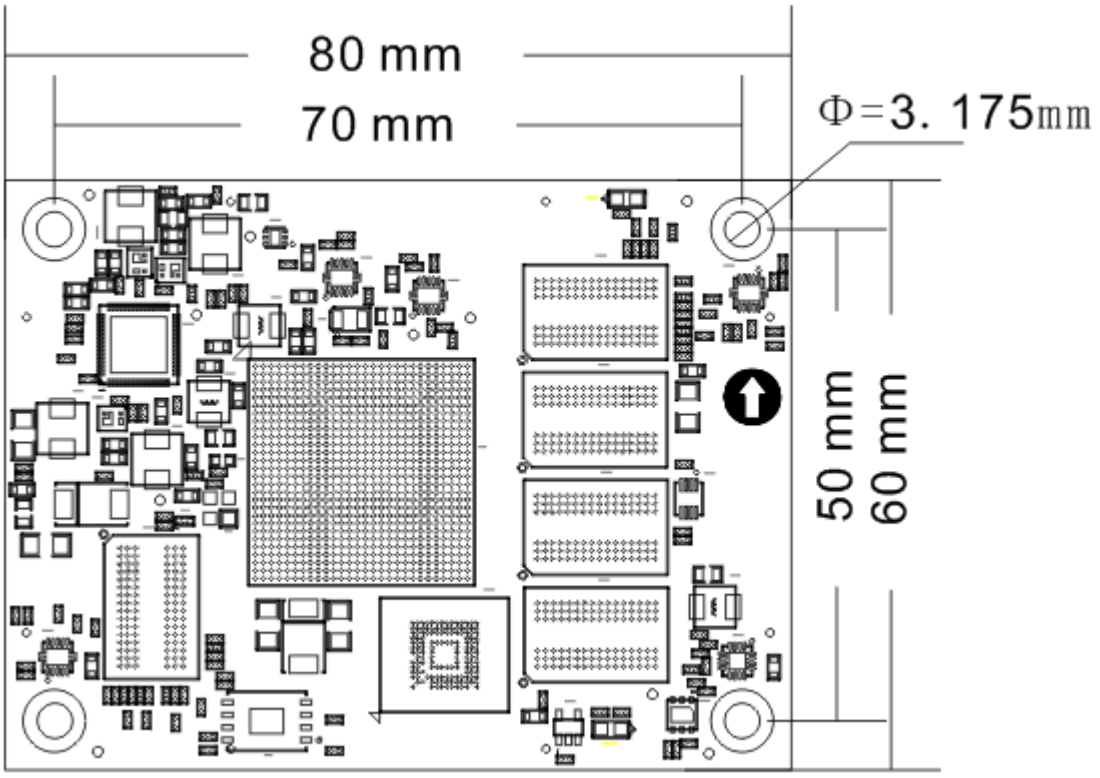
1.8

Power supply

The power supply voltage of the ACU4EV core board is + 12V, and power is supplied to the core board by connecting the baseplate. All the power required for the XCZU4EV chip is generated on the core board through a PMIC chip TPS6508641. Please refer to the power chip manual for the design of the TPS6508641 power supply. The design diagram is as follows:



In addition, the VCCIO power supply of the BANK65 and BANK66 of the XCZU4EV chips is provided by the baseplate, which is convenient for users to modify, but the maximum power supply cannot exceed 1.8V.


80 mm
70 mm
Φ=3. 175mm
50 mm
60 mm

79	PS_MIO49	M18	80	PS_MIO75	A19
81	PS_MIO34	L17	82	PS_MIO70	C19
83	GND	-	84	GND	-

85	PS_MIO26	L15	86	PS_MIO43	K19
87	PS_MIO24	AB19	88	PS_MIO51	L21
89	GND	-	90	GND	-
91	PS_MIO25	AB21	92	PS_MIO42	L18
93	-	-	94	PS_MIO33	L16
95	GND	-	96	GND	-
97	-	-	98	-	-

99	VCCO_65	-	100	VCCO_66	-
101	VCCO_65	-	102	VCCO_66	-
103	VCCO_65	-	104	VCCO_66	-
105	GND	-	106	GND	-
107	+12V	-	108	+12V	-
109	+12V	-	110	+12V	-
111	+12V	-	112	+12V	-

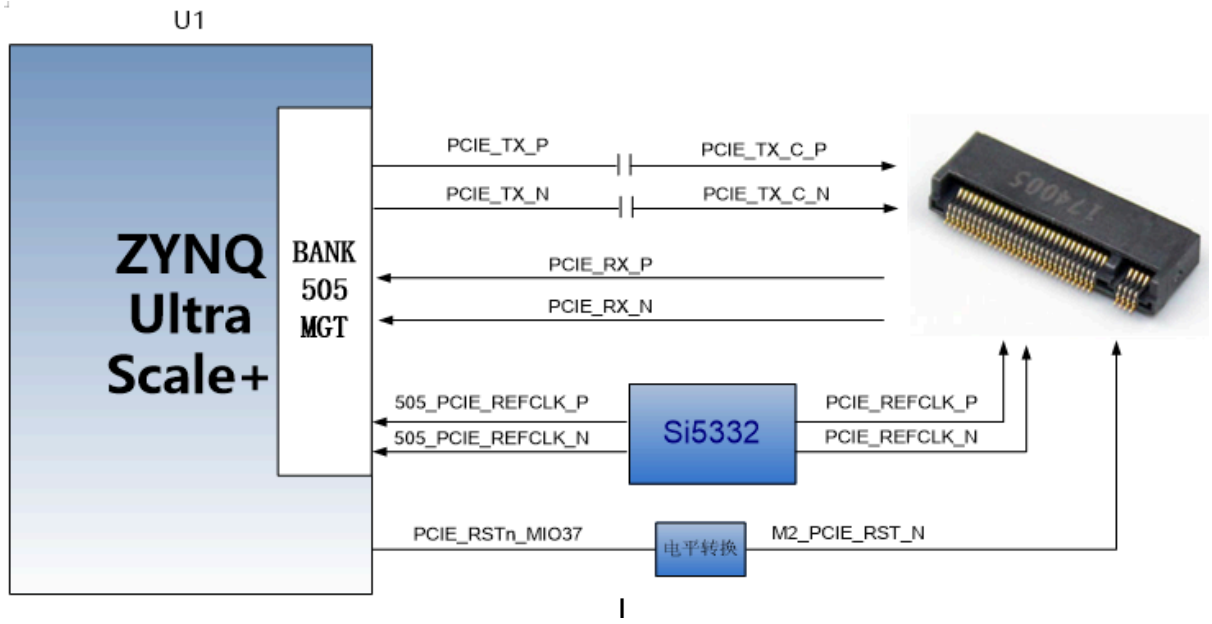
113	+12V	-	114	+12V	-
115	+12V	-	116	+12V	-
117	+12V	-	118	+12V	-
119	+12V	-	120	+12V	-

2.2

M.2 interface

The AXU4EV-P development board is equipped with a PCIE x1 standard M.2 interface for connecting to an M.2 SSD with a communication speed of up to 6Gbps. The M.2 interface uses an M key slot and only supports PCI-E, not SATA. When users choose an SSD, they need to choose a PCIE type SSD.

The PCIE signal is directly connected to the ZU4EV's BANK505 PS MGT transceiver. The TX and RX signals are connected to the MGT's LANE1 in a differential signal manner. The PCIE clock is provided by the Si5332 chip, with a frequency of 100Mhz. The schematic diagram of the M.2 circuit design is shown in the following figure 3-2-1:



Signal name	Pin name	Pin number	Remarks

CIE_TX_P	05_TX0_P	25	PCIE data transmission positive
CIE _TX_N	05_TX0_N	26	PCIE data transmission negative
CIE _RX_P	05_RX0_P	27	PCIE data reception positive

'CIE _RX_N	05_RX0_N	28	PCIE data reception negative
05_PCIE_REFCLK_P	05_CLK0_P	23	PCIE reference clock positive
05_PCIE_REFCLK_N	05_CLK0_N	24	PCIE reference clock negative

'CIE_RSTn_MIO37	S_MIO37_501	17	PCIE reset signal

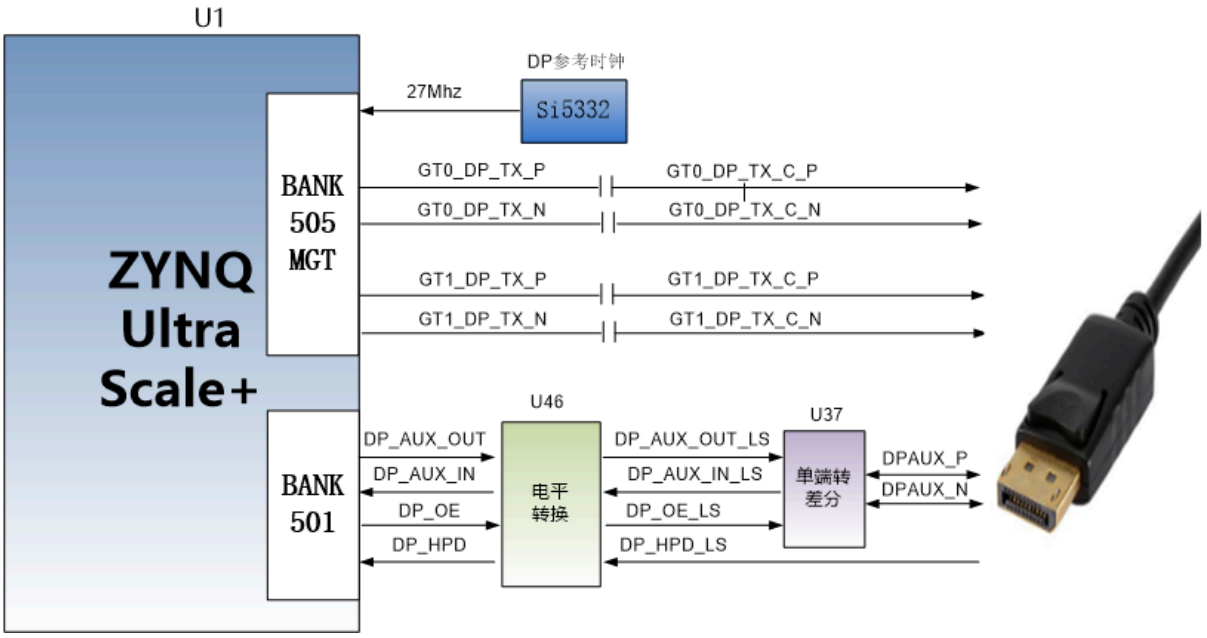
2.3

DP display interface

The AXU4EV-P development board comes with a standard DisplayPort output display interface for displaying video images. The interface supports the VESA DisplayPort V1.2a output standard, up to 4K x 2K@30Fps output, and supports Y-only, YCbCr444, YCbCr422, YCbCr420 and RGB video formats. Each color supports 6, 8, 10, or 12 bits.

The DisplayPort data transmission channel is directly driven by ZU4EV's BANK505 PS MGT output. The LANE2 and LANE3 TX signals of the MGT are connected to the DP connector in a differential signal manner. The DisplayPort auxiliary channel is

connected to the MIO pin of the PS. The schematic diagram of the DP output interface design is shown in the following figure 3-3-1:



3-3-1 DP interface design diagram

DisplayPort interface ZYNQ pin allocation is as follows:

Signal name	ZYNQ pin name	ZYNQ pin number	Remarks

GT0_DP_TX_P	505_TX3_P	B23	DP data low transmission positive
GT0_DP_TX_N	505_TX3_N	B24	DP data low transmission negative
GT1_DP_TX_P	505_TX2_P	C25	DP data high-level transmission positive

GT1_DP_TX_N	505_TX2_N	C26	DP data high-level transmission negative
505_CLK1_P	505_CLK2_P	C21	DP reference clock is positive
505_CLK1_N	505_CLK2_N	C22	DP reference clock negative

DP_AUX_OUT	PS_MIO27	J15	DP auxiliary data output
DP_AUX_IN	PS_MIO30	F16	DP auxiliary data input
DP_OE	PS_MIO29	G16	DP auxiliary data output enable

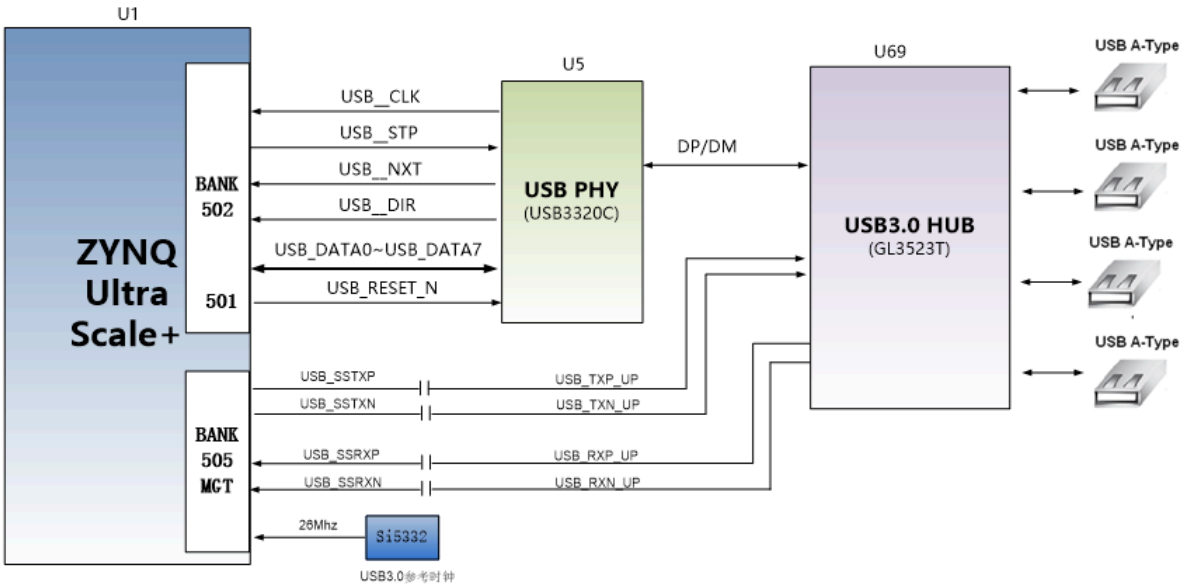
DP_HPD	PS_MIO28	K15	DP insertion signal detection

2.4

USB 3.0 interface

The AXU4EV-P expansion board has 4 USB3.0 interfaces, supports HOST mode, and has a data transfer speed of up to 5.0Gb/s. USB3.0 is connected through the PIPE3 interface, and USB2.0 is connected to the external USB3320C chip through the ULPI interface, achieving high-speed data communication between USB3.0 and USB2.0.

The USB interface is a flat USB interface (USB Type A), which is convenient for users to connect different USB slave peripherals (such as USB mouse, keyboard or USB flash drive) at the same time. The schematic diagram of the USB 3.0 connection is shown in 3-4-1:


U1
ZYNQ
Ultra
Scale+
BANK
502
501
BANK
505
MGT
U5
USB PHY
(USB3320C)
USB_CLK
USB_STP
USB_NXT
USB_DIR
USB_DATA0~USB_DATA7
USB_RESET_N
DP/DM
U69
USB3.0 HUB
(GL3523T)
USB A-Type
USB A-Type
USB A-Type
USB A-Type
USB_SSTXP
USB_SSTXN
USB_SSRXP
USB_SSRXN
USB_TXP_UP
USB_TXN_UP
USB_RXP_UP
USB_RXN_UP
26Mhz
Si5332
USB3.0参考时钟

USB_NXT	PS_MIO55	B16	USB 2.0 next data signal

USB_RESET_N	PS_MIO31	H16	USB2.0 reset signal

2.5

Gigabit Ethernet interface

The AXU4EV-P expansion board has two Gigabit Ethernet interfaces, one connected to the PS end and the other connected to the PL end. The GPHY chip uses Micrel's KSZ9031RNX Ethernet PHY chip to provide users with network communication services. The KSZ9031RNX chip supports 10/100/1000 Mbps network transmission rates and communicates with the MAC layer of the ZU4EV system through the RGMII interface. The KSZ9031RNX supports MDI/MDX Self-Adaptation, various speed Self-Adaptation, Master/Slave Self-Adaptation, and MDIO bus for PHY register management.

The KSZ9031RNX will detect the level status of some specific IO to determine its operating mode. Table 3-5-1 describes the default setting information of the GPHY chip after power-on.

Configure pin	Explanation	Configuration value
PHYAD[2:0]	PHY address for MDIO/MDC mode	PHY Address 011
CLK125_EN	Enable 125Mhz clock output selection	Enable

LED_MODE	LED light mode configuration	Single LED light mode
MODE0~MODE3	Link Self-Adaptation and Full-Duplex Configuration	10/100/1000 Self-Adaptation, compatible with full-duplex and half-duplex

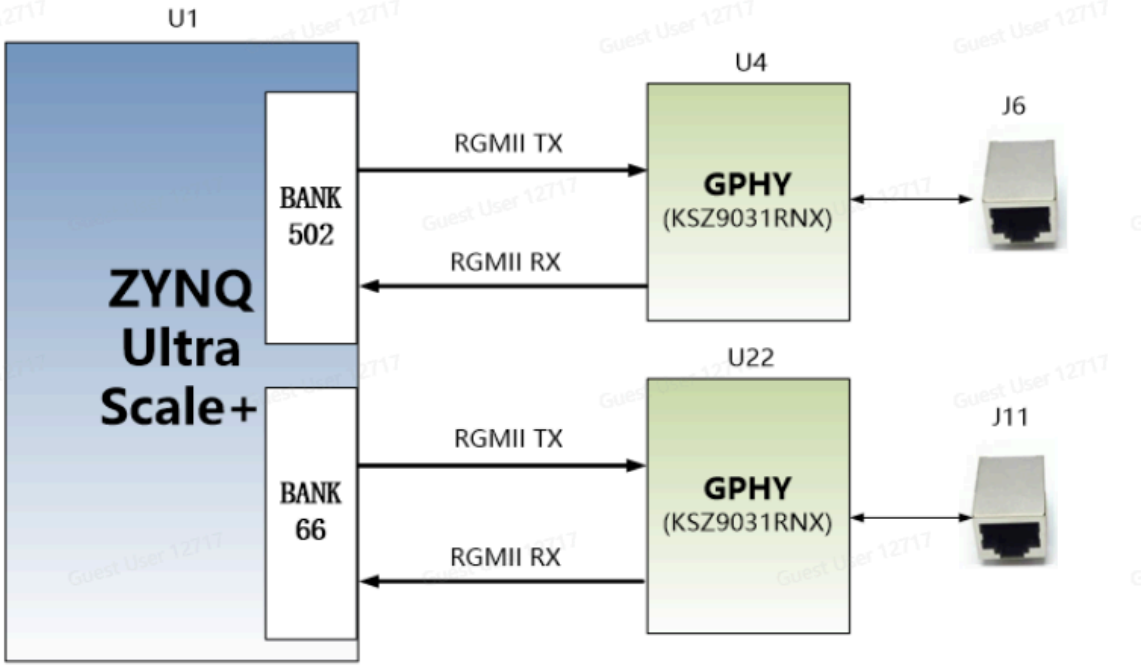
Table 3-5-1 PHY chip default configuration values

When the network is connected to Gigabit Ethernet, the data transmission between ZYNQ and PHY chip KSZ9031RNX is communicated through the RGMII bus, with a

transmission clock of 125Mhz, and the data is sampled at the rising edge and falling sample of the clock.

When the network is connected to 100Mbps Ethernet, the data transmission between ZYNQ and PHY chip KSZ9031RNX is communicated through the RMII bus, and the transmission clock is 25Mhz. The data is sampled at the rising edge and falling sample of the clock.

Figure 3-5-1 is a schematic diagram of ZYNQ Ethernet PHY chip connection.



			Ethernet 2 RGMII receive clock
PHY2_RXD0	B66_L19_N	A5	thernet 2 receives data Bit0
PHY2_RXD1	B66_L19_P	B5	thernet 2 receives data Bit1

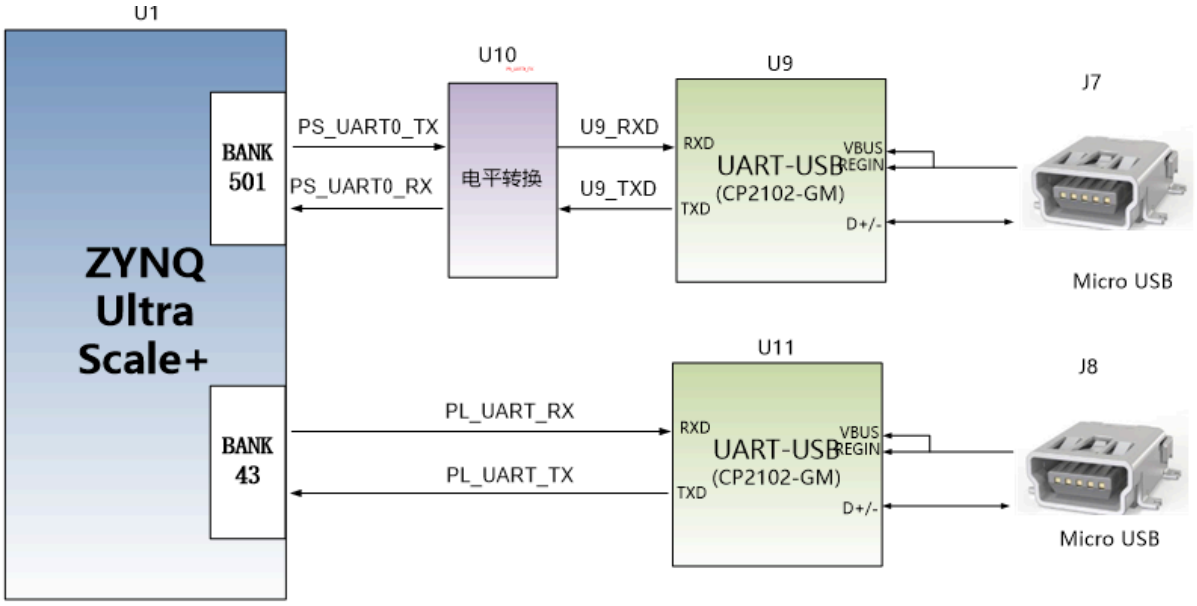
PHY2_RXD2	B66_L17_P	F8	thernet 2 receives data Bit2
PHY2_RXD3	B66_L24_P	C9	thernet 2 receives data Bit3
PHY2_RXCTL	B66_L22_N	B8	thernet 2 receives data valid signal

PHY2_MDC	B66_L21_N	A6	Ethernet 2 MDIO management clock
PHY2_MDIO	B66_L22_P	C8	Ethernet 2 MDIO manages data
PHY2_RESET	B66_L14_N	D5	Ethernet 2 reset signal

2.6

USB Uart interface

The AXU4EV-P expansion board is equipped with two Uart to USB interfaces, one connected to the PS end and one connected to the PL end. The conversion chip uses the USB-UAR chip of Silicon Labs CP2102GM, and the USB interface uses the MINI USB interface, which can be connected to the USB port of the PC for serial data communication with a USB cable. The schematic diagram of the USB Uart circuit design is shown in the following figure.



3-6-1 USB to Serial

ZYNQ pin allocation for USB to serial port:

Signal name	Pin name	Pin number	Remarks
PS_UART0_TX	PS_MIO43	K19	PS Uart data output

PS_UART0_RX	PS_MIO42	L18	PS Uart Data Input
PL_UART_TX	B43_L9_P	AA11	PL Uart data output
PL_UART_RX	B43_L9_N	AA10	PL Uart Data Input

2.7

SD card slot

The AXU4EV-P expansion board includes a Micro SD card interface to provide users with access to SD card memory for storing BOOT programs for ZU4EV chips, Linux operating system kernels, file systems, and other user data files.

The SDIO signal is connected to the IO signal of the PS BANK501 of ZU4EV, because the VCCIO of 501 is set to 1.8V, but the data level of the SD card is 3.3V. Here we connect it through the TXS02612 level converter. The schematic diagram of the PS and SD card connectors of ZU4EV is shown in Figure 3-7-1.

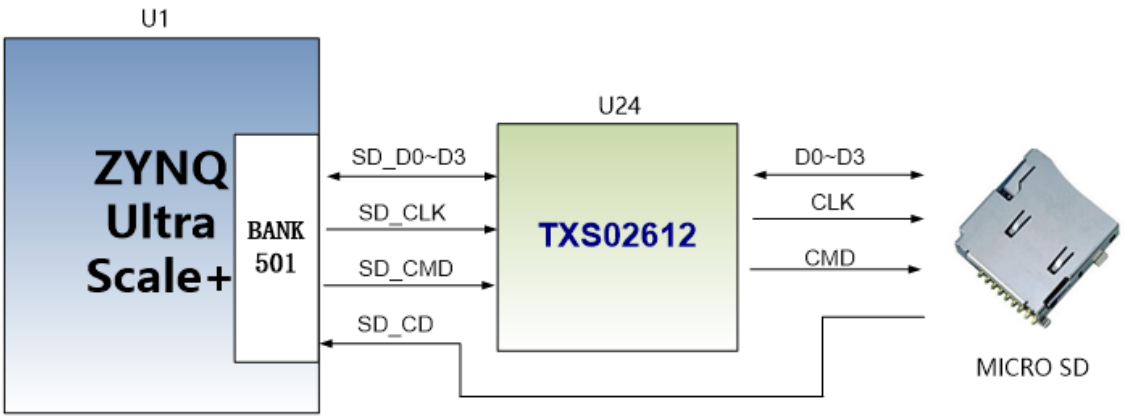


Figure 3-7-1 SD card connection diagram

SD card slot pin distribution

Signal name	Pin name	Pin number	Remarks
SD_CLK	PS_MIO51	l21	SD clock signal
SD_CMD	PS_MIO50	M19	SD command signal
SD_D0	PS_MIO46	L20	SD Data0

SD_D1	PS_MIO47	H21	SD Data1
SD_D2	PS_MIO48	J21	SD Data Data2
SD_D3	PS_MIO49	M18	SD Data3

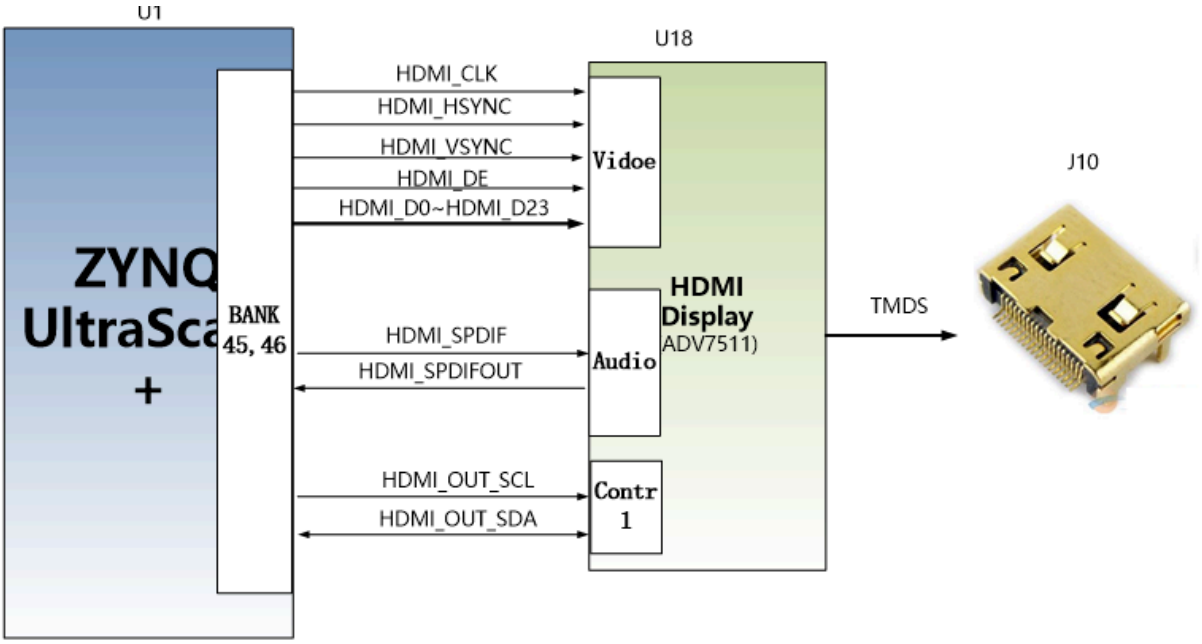
SD_CD	PS_MIO45	K20	D card detection signal

2.8

HDMI output interface

The implementation of HDMI output interface is the use of ANALOG DEVICE ADV7511 HDMI (DVI) encoding chip, the highest support 1080P@60Hz output, support 3D output.

Among them, the video digital interface, audio digital interface, and I2C configuration interface of ADV7511 are connected to the BANK45,46 IO of the ZU4EV PL part. The ZYNQ UltraScale + system initializes and controls ADV7511 through the I2C pin. The hardware connection diagram of ADV7511 chip and ZYNQ UltraScale + is shown in the following figure 3-8-1:


U1
ZYNQ
UltraSca
+
BANK
45, 46
U18
HDMI_CLK
HDMI_HSYNC
HDMI_VSYNC
HDMI_DE
HDMI_D0~HDMI_D23
Vidoe
HDMI
Display
ADV7511)
HDMI_SPDIF
HDMI_SPDIFOUT
Audio
HDMI_OUT_SCL
HDMI_OUT_SDA
Contr
1
TMDS
J10

HDMI_D17	B35_L15_P	F17	HDMI video signal data 17

HDMI_D18	B35_L7_N	G16	HDMI video signal data 18
HDMI_D19	B35_L7_P	G17	HDMI video signal data 19
HDMI_D20	B35_L6_N	H16	HDMI video signal data 20

HDMI_D21	B35_L6_P	J16	HDMI video signal data 21
HDMI_D22	B35_L17_P	C17	HDMI video signal data 22
HDMI_D23	B35_L17_N	B16	HDMI video signal data 23

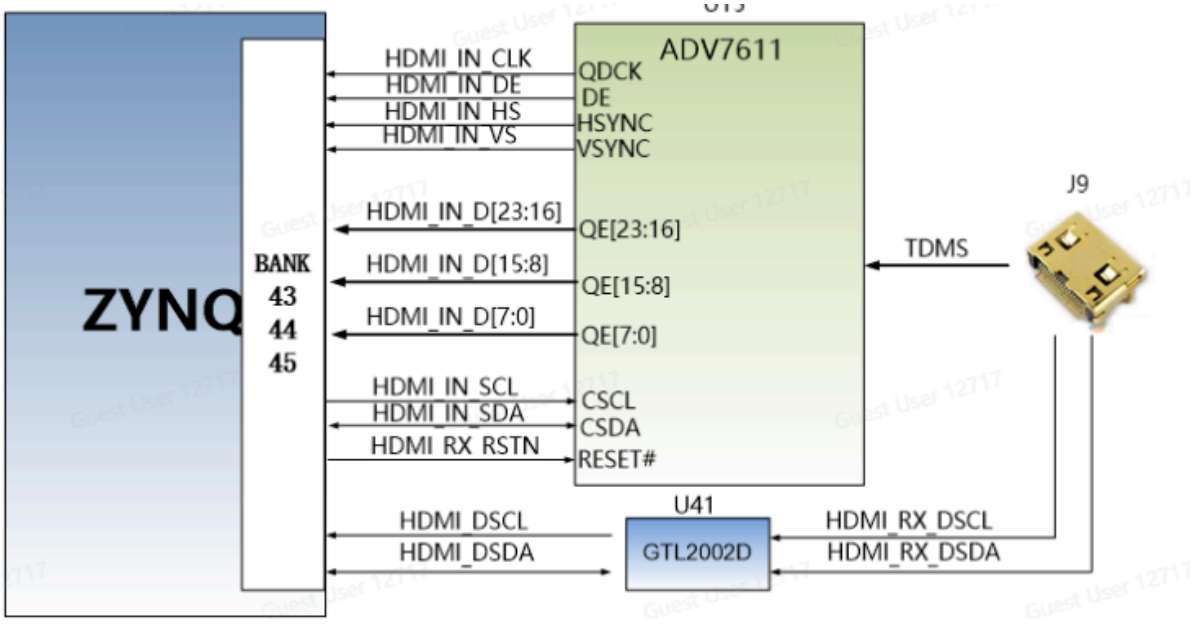
HDMI_SCL	B35_L13_N	E15	HDMI IIC control clock
HDMI _SDA	B35_L13_P	E16	HDMI IIC control data

2.9

HDMI input interface

HDMI input interface We use ANALOG DEVICE ADV7611 decoding chip, the highest support 1080P@60Hz input, support different formats of data output.

Among them, the IIC configuration interface of ADV7611 is also connected to the IO of ZU4EV PL, and ZYNQ initializes and controls ADV7611 through the programming of I2C bus. The hardware connection of HDMI input interface is shown in Figure 3-9-1.



IDMI_IN_D15	43_L10_P	/10	DMI input video signal ata 16
IDMI_IN_D16	44_L11_N	/11	DMI input video signal ata 17
IDMI_IN_D17	44_L11_P	/12	DMI input video signal ata 18

IDMI_IN_D18	44_L9_N	/13	DMI input video signal ata 19
IDMI_IN_D19	44_L9_P	/14	DMI input video signal ata 20
IDMI_IN_D20	44_L8_P	B15	DMI input video signal ata 21

IDMI_IN_D21	44_L8_N	B14	DMI input video signal ata 22
IDMI_IN_D22	43_L2_N	G11	DMI input video signal ata 23
IDMI_IN_SDA	43_L5_P	E12	DV7611 IIC control lock

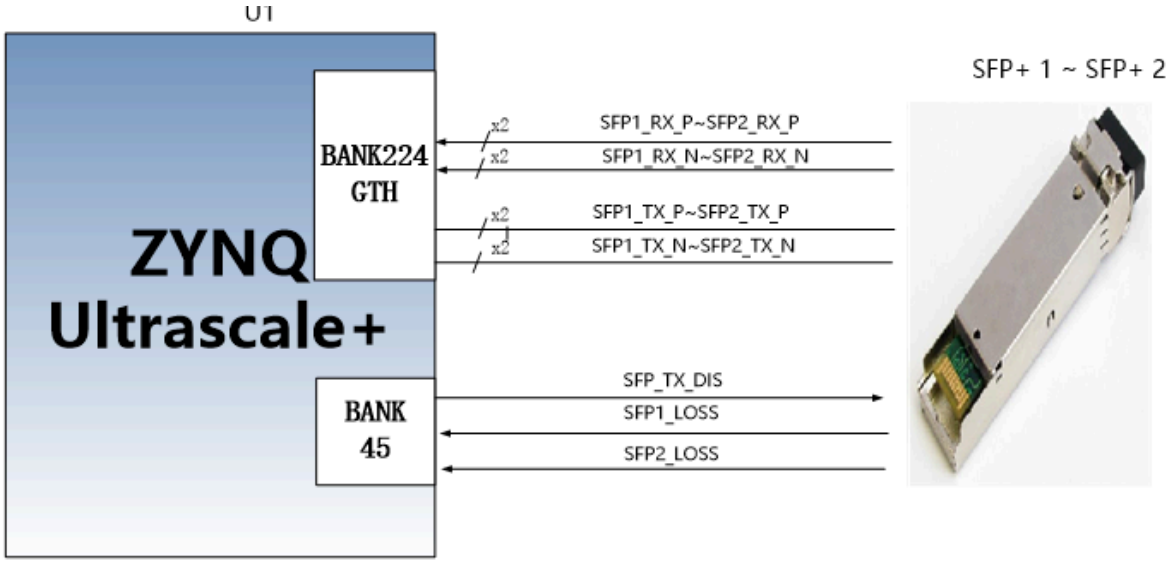
IDMI_IN_SCL	43_L5_N	F12	DV7611 IIC control ata
IDMI_DSCL	45_L5_N	10	EDID read IIC clock
IDMI_DSDA	45_L5_P	11	EDID reads IIC data

2.10

Fiber optic interface

The AXU4EV-P expansion board has two fiber optic interfaces. Users can purchase SFP optical modules (1.25G, 2.5G, 10G optical modules on the market) and insert them into these two fiber optic interfaces for fiber optic data communication. The two fiber optic interfaces are respectively connected to the two RX/TX channels of the GTH transceiver of ZYNQ's BANK224. The data rate of each TX transmission and RX reception is as high as 12.5Gb/s. The reference clock of the GTH transceiver is provided by the 125M differential clock of the core board.

The ZYNQ Ultrascale + and fiber design diagram is shown in the following figure 3-10-1:



2-Channel fiber optic interface ZYNQ pin allocation is as follows:

Signal name	ZYNQ pin name	ZYNQ pin number	Remarks
SFP1_TX_N	224_TX0_N	W3	Optical Module 1 Data Transmission Negative

SFP1_TX_P	224_TX0_P	W4	Optical Module 1 Data Transmission Positive
SFP1_RX_N	224_RX0_N	Y1	Optical module 1 receives negative data
SFP1_RX_P	224_RX0_P	Y2	Optical Module 1 Data Reception Positive

SFP2_TX_N	224_TX1_N	U3	Optical Module 2 Data Transmission Negative
SFP2_TX_P	224_TX1_P	U4	Optical Module 2 Data Transmission Positive
SFP2_RX_N	224_RX1_N	V1	Optical module 2 receives negative data

SFP2_RX_P	224_RX1_P	V2	Optical module 2 receives data positively
SFP_TX_DIS	B45_L12_P	D12	Optical module light emission prohibition, high efficiency
SFP1_LOSS	B45_L9_N	B10	

			Optical module 1 receives LOSS detection signal
SFP2_LOSS	B45_L9_P	C11	Optical module 2 receives LOSS detection signal

2.11

PCIe slot

The AXU4EV-P expansion board has a PCIe x2 interface, and two pairs of transceivers are connected to the golden finger of PCIEx2, which can achieve PCIE 3.0 (compatible with 2.0) data communication.

The transceiver signal of the PCIe interface is directly connected to the GTH transceiver of the ZYNQ BANK224, and the single-channel communication rate can reach up to 8G bit bandwidth.

The design diagram of the PCIe interface of the development board is shown in Figure 3-11-1, where the TX transmission signal is connected in AC coupling mode.

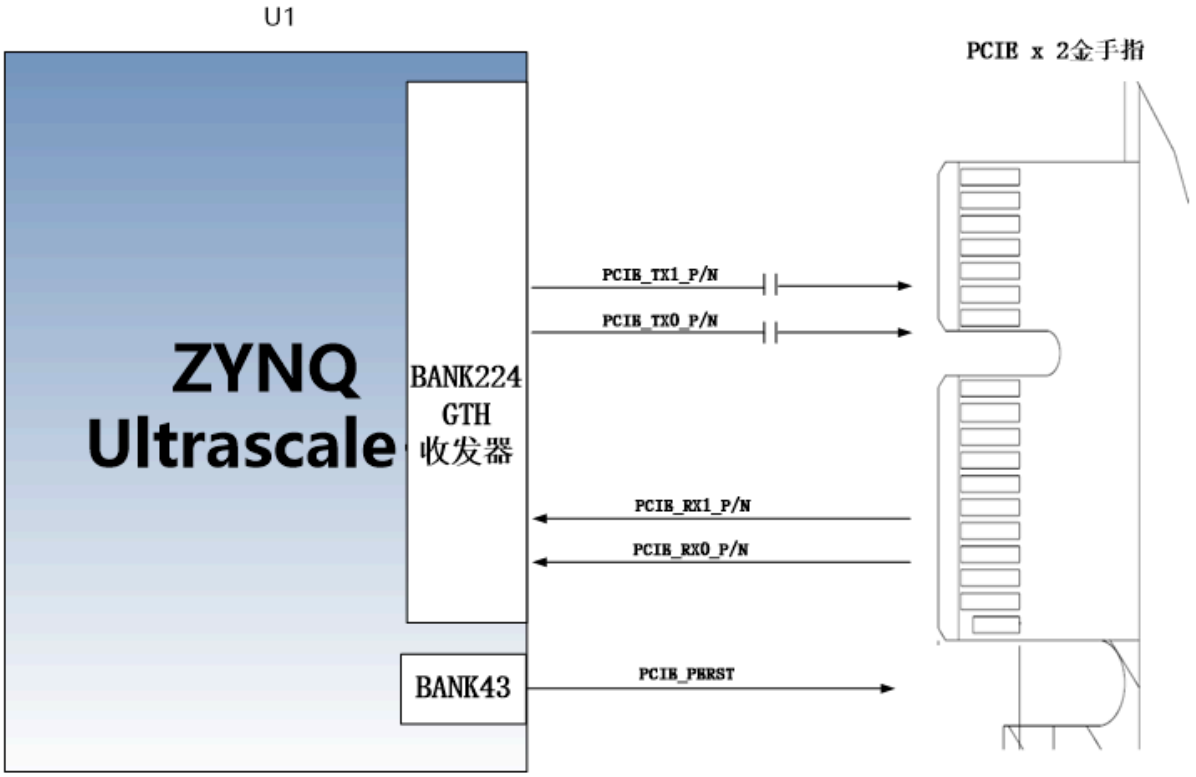


Figure 3-11-1 PCIe slot design schematic

PCIe x2 interface ZYNQ pin allocation is as follows:

Signal name	ZYNQ pin name	ZYNQ pin number	Remarks
PCIE_RX0_N	224_RX2_N	T1	PCIE channel 0 data reception negative
PCIE_RX0_P	224_RX2_P	T2	PCIE channel 0 data reception positive

PCIE_RX1_N	224_RX3_N	P1	PCIE channel 1 data reception negative
PCIE_RX1_P	224_RX3_P	P2	PCIE channel 1 data reception positive
PCIE_TX0_N	224_TX2_N	R3	PCIE channel 0 data transmission negative

PCIE_TX0_P	224_TX2_P	R4	PCIE channel 0 data transmission positive
PCIE_TX1_N	224_TX3_N	N3	PCIE channel 1 data transmission negative
PCIE_TX1_P	224_TX3_P	N4	PCIE channel 1 data transmission positive

PCIE_PERST	B43_L8_N	AC11	Reset signal for PCIE board

2.12

CAN communication interface

The AXU4EV-P expansion board has two CAN communication interfaces, which are connected to the MIO interface of the PS system side BANK501. The CAN transceiver chip uses TI's SN65HVD232C chip to provide users with CAN communication services.

Figure 3-12-1 is a schematic diagram of the connection of the PS-side CAN transceiver chip

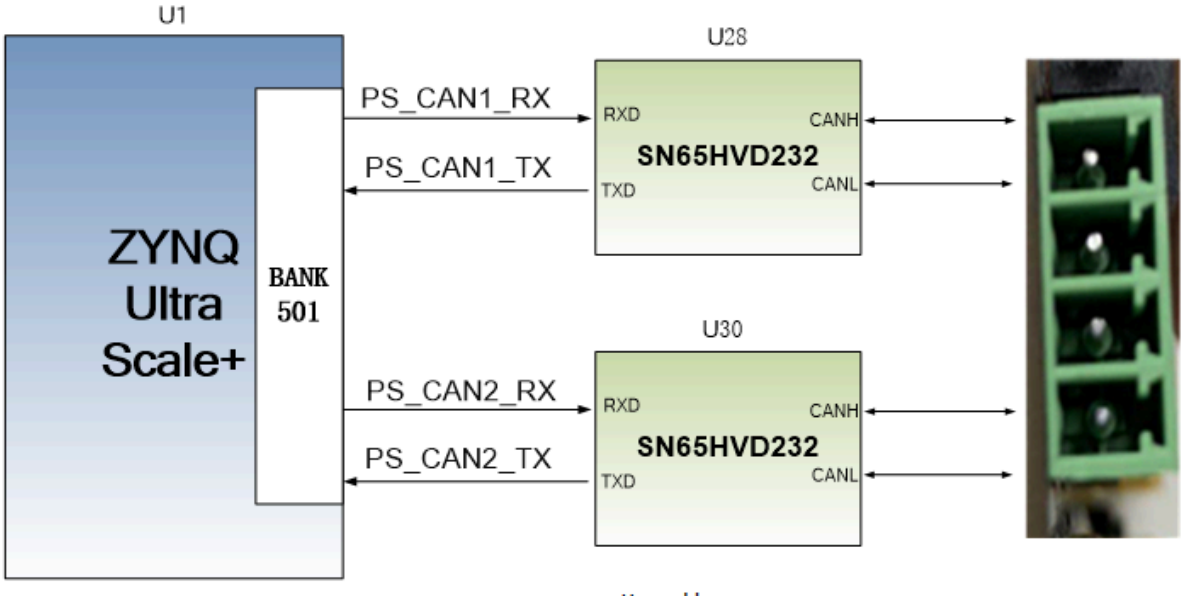


Figure 3-12-1 connection schematic diagram of PS side CAN transceiver chip

CAN communication pin allocation is as follows:

Signal name	Pin name	Pin number	Remarks

PS_CAN1_TX	PS_MIO32	J16	CAN1 sender
PS_CAN1_RX	PS_MIO33	L16	CAN1 receiver
PS_CAN2_TX	PS_MIO39	H19	CAN2 sender
PS_CAN2_RX	PS_MIO38	H18	CAN2 receiver

2.13

485 communication interface

The AXU4EV-P expansion board has two 485 communication interfaces, and the 485 communication port is connected to the IO interface of the PL end BANK43 and 45. The 485 transceiver chip uses MAX3485 chip from MAXIM company to provide users with 485 communication services.

Figure 3-13-1 is a schematic diagram of the connection of the PL terminal 485 transceiver chip

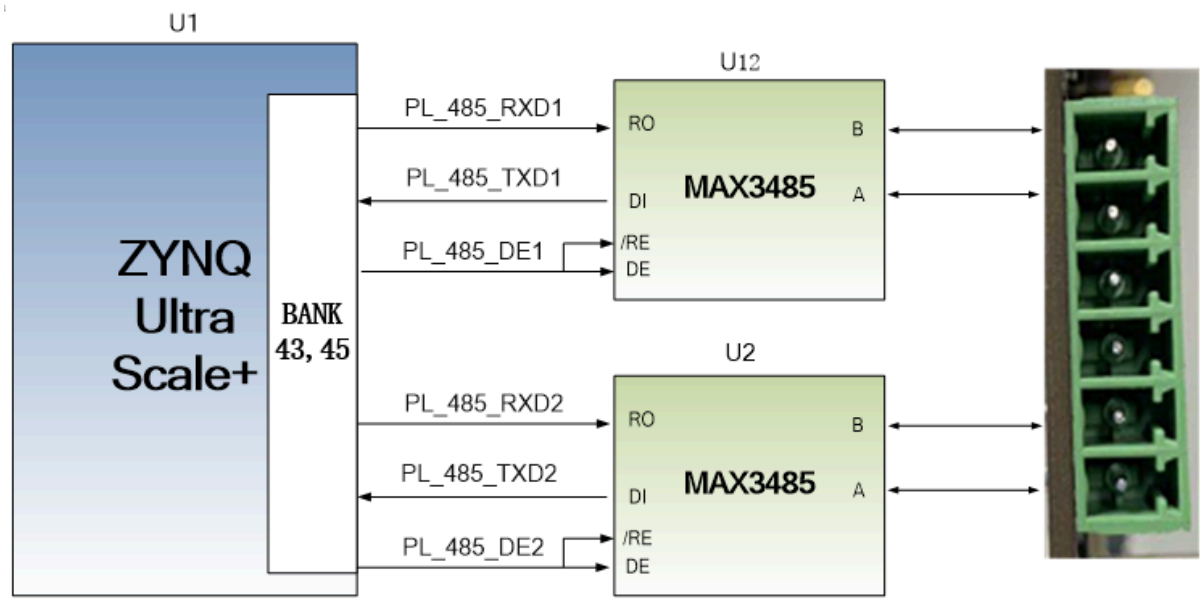


Figure 3-13-1 connection diagram of PL terminal 485 communication

RS485 communication pin allocation is as follows:

Signal name	Pin name	Pin number	Remarks
PL_485_TXD1	B43_L1_P	AG10	First channel 485 transmitter
PL_485_RXD1	B43_L1_N	AH10	

			First channel 485 receiving end
PL_485_DE1	B45_L10_N	A10	First Route 485 Send Enable
PL_485_TXD2	B43_L3_N	AH11	Second channel 485 transmitter

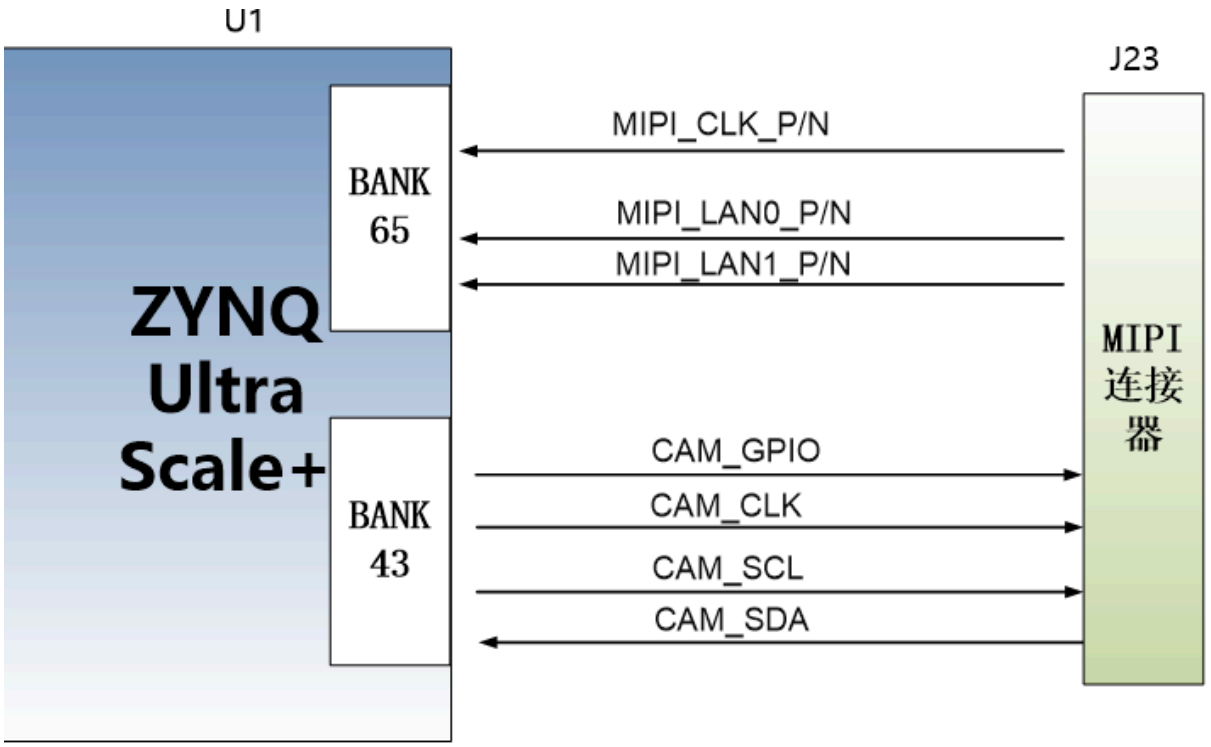
PL_485_RXD2	B43_L3_P	AH12	he second channel 485 receiving end
PL_485_DE2	B45_L10_P	B11	econd Route 485 Send Enable

2.14

MIPI interface

The expansion board includes a MIPI camera interface that can be used to connect our MIPI OV5640 camera module (AN5641). The MIPI interface has a 15PIN FPC connector for 2 LANE data and 1 pair of clocks, connected to the differential IO pin of BANK65, with a level standard of 1.2V (J43's jump cap needs to be connected to PIN2

and PIN3 pins); other control signals are connected to BANK43's IO, with a level standard of 3.3V.



MIPI interface pin assignment

Signal name	ZYNQ pin name	ZYNQ pin number	Remarks

MIPI_CLK_P	B65_L1_P	W8	MIPI input clock is positive
MIPI_CLK_N	B65_L1_N	Y8	MIPI input clock negative
MIPI _ LAN0 _ P	B65_L2_P	U9	MIPI input data LANE0 positive

MIPI_LAN0_N	B65_L2_N	V9	MIPI input data LANE0 negative
MIPI _ LAN1 _ P	B65_L3_P	U8	MIPI input data LANE1 positive
MIPI_LAN1_N	B65_L3_N	V8	MIPI input data LANE1 negative

CAM_GPIO	B43_L4_P	AE10	GPIO control of the camera
CAM_CLK	B43_L4_N	AF10	Clock input of the camera
CAM_SCL	B43_L11_P	Y9	Camera's I2C clock

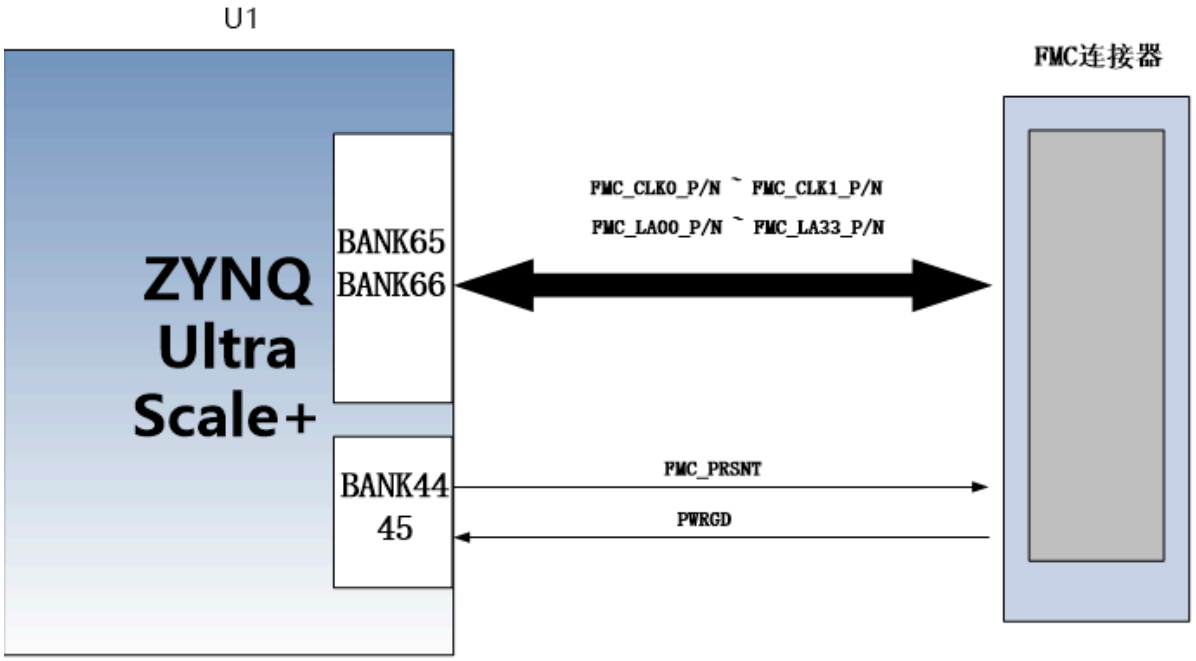
CAM_SDA	B43_L11_N	AA8	Camera I2C data

2.15

FMC connector

The AXU4EV-P expansion board comes with a standard FMC LPC expansion port, which can be connected to various FMC modules (HDMI input/output module, binocular camera module, high-speed AD module, etc.) from XILINX or our Black Gold. The FMC expansion port contains 36 pairs of differential IO signals.

The 36 pairs of differential signals of the FMC expansion port are connected to the IO of the BANK65,66 of the ZYNQ Ultrascale + chip, with a level standard of 1.8V. The differential signals support LVDS data communication. The schematic diagram of the ZYNQ Ultrascale + and FMC connectors is shown in Figure 3-15-1.


U1
FMC连接器
ZYNQ
Ultra
Scale+
BANK65
BANK66
BANK44
45
FMC_CLK0_P/N ~ FMC_CLK1_P/N
FMC_LA00_P/N ~ FMC_LA33_P/N
FMC_PRSNT
PWRGD

MC_LA30_P	66_L13_P	D7	MC reference 30th hannel data P
MC_LA30_N	66_L13_N	D6	MC reference 30th ata N

MC_LA31_P	66_L9_P	B3	MC reference 31st ata P
MC_LA31_N	66_L9_N	A3	MC reference 31st ata N
MC_LA32_P	65_L24_P	H9	MC reference 32nd ata P

MC_LA32_N	65_L24_N	H8	MC reference 32nd ata N
MC_LA33_P	66_L8_P	A2	MC reference 33rd ata P
MC_LA33_N	66_L8_N	A1	MC reference 33rd ata N

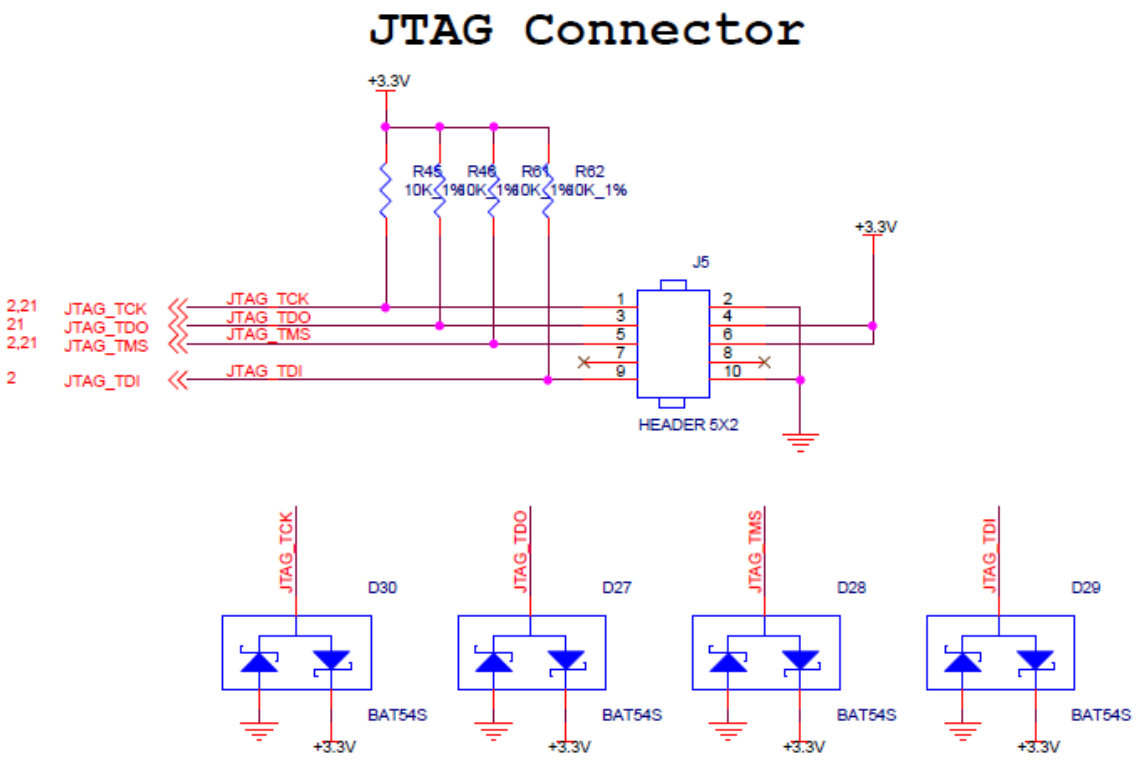
MC_PRSNT	45_L12_N	12	MC module presence ignal
'WRGD	44_L2_N	H14	MC Power Goods ignal

2.16

JTAG debugging port

A JTAG interface is reserved on the AXU4EV-P expansion board for downloading ZYNQ UltraScale + programs or solidifying programs to FLASH. In order to prevent damage to the ZYNQ UltraScale + chip caused by live plugging and unplugging, we

added a protective diode to the JTAG signal to ensure that the voltage of the signal is within the range accepted by the FPGA, avoiding damage to the ZYNQ UltraScale + chip.

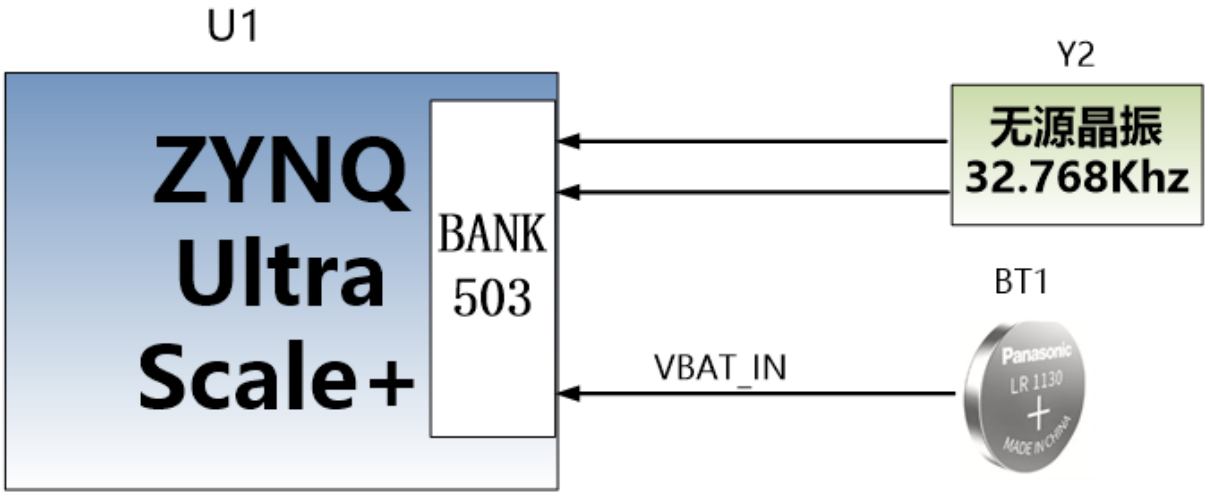


2.17

RTC real-time clock

The ZU4EV chip has an internal RTC real-time clock function, which includes year, month, day, hour, minute, second, and week timing functions. An external passive clock of 32.768KHz needs to be connected to provide an accurate clock source to the internal clock circuit, so that the RTC can accurately provide clock information. At the same time, in order to ensure that the real-time clock can run normally after the product is powered down, an additional battery is generally needed to power the clock chip. The

BT1 on the development board is a 1.5V button battery (model LR1130, voltage is 1.5V). When the system drops the battery, the button battery can also power the RTC system and provide continuous time information. Figure 3-12-1 is the schematic diagram of the RTC real-time clock



2.18

EEPROM and temperature sensors

The AXU4EV-P development board is equipped with an EEPROM, model 24LC04, with a capacity of 4Kbit (2 * 256 * 8bit), which is connected to the PS end for communication through the IIC bus. In addition, the board also has a high-precision, low-power, digital temperature sensor chip, model LM75 from ON Semiconductor, with a temperature accuracy of 0.5 degrees. The EEPROM and temperature sensor are mounted on the Bank500 MIO of ZYNQ UltraScale + through the I2C bus. Figure 3-18-1 is the schematic diagram of the EEPROM and temperature sensor

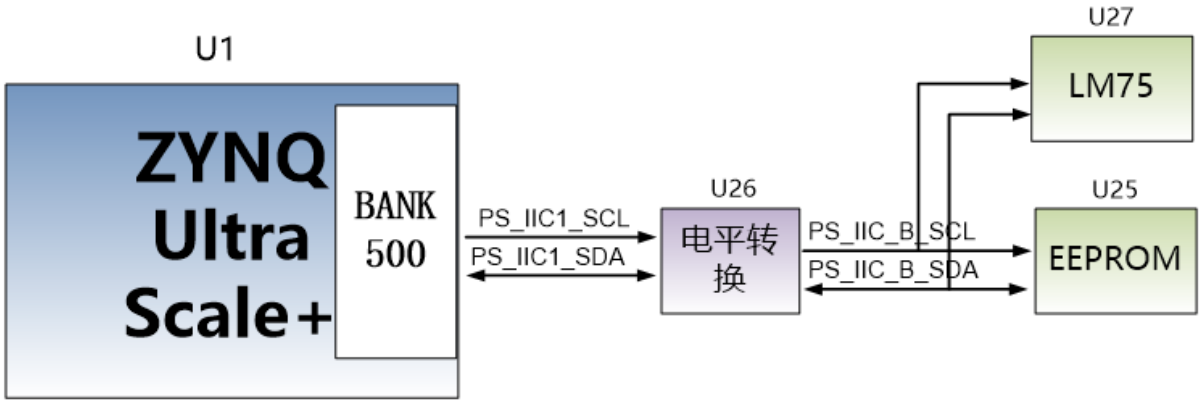


Figure 3-18-1 Schematic of EEPROM and sensor

EEPROM communication pin allocation is as follows:

Signal name	Pin name	Pin number	Remarks

PS_IIC1_SCL	PS_MIO24	AB19	I2C clock signal
PS_IIC1_SDA	PS_MIO25	AB21	I2C data signal

2.19

LED lights

The AXU4EV-P expansion board has 3 Light Emitting Diode LEDs, including 1 power indicator, 1 PS control indicator, and 1 PL control indicator. Users can control the on and off through the program. When the IO voltage connected to the user LED is low, the

user LED will turn off. When the IO voltage connected to the user LED is high, the user LED will light up. The hardware connection diagram of the user LED is shown in Figure 3-19-1.

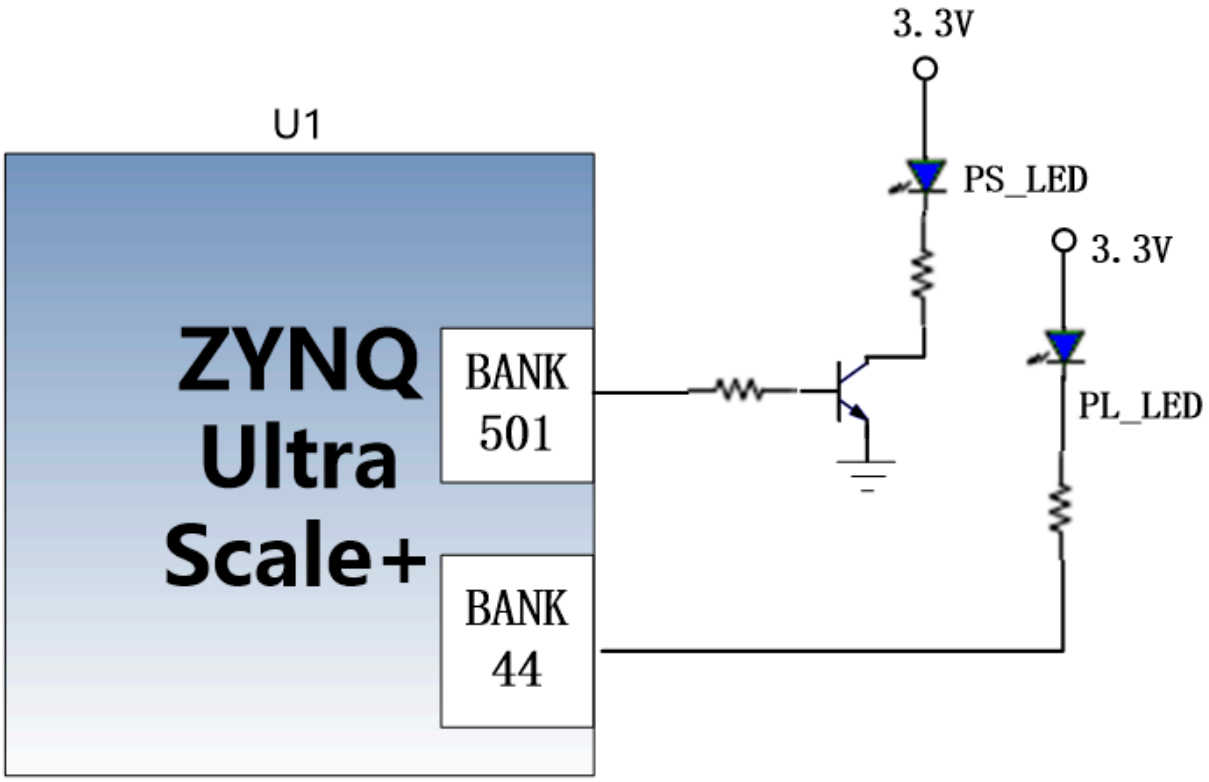


Figure 3-19-1 user LED lamp hardware connection diagram

1.

Pin allocation for user LED lights

Signal name	Pin name	Pin number	Remarks
'S_LED1	S_MIO40	18	ser PS LED lamp
'L_LED1	44_L1_P	E15	ser PL LED lamp

2.20

Button

The AXU4EV-P expansion board has one reset button RESET and two user buttons. The reset signal is connected to the reset chip input of the core board, and the user can use this reset button to reset the ZYNQ system. One user button is connected to the MIO of PS, and the other is connected to the IO of PL. Both the reset button and the user button are active at a low level. The connection diagram of the user button is shown in Figure 3-20-1.

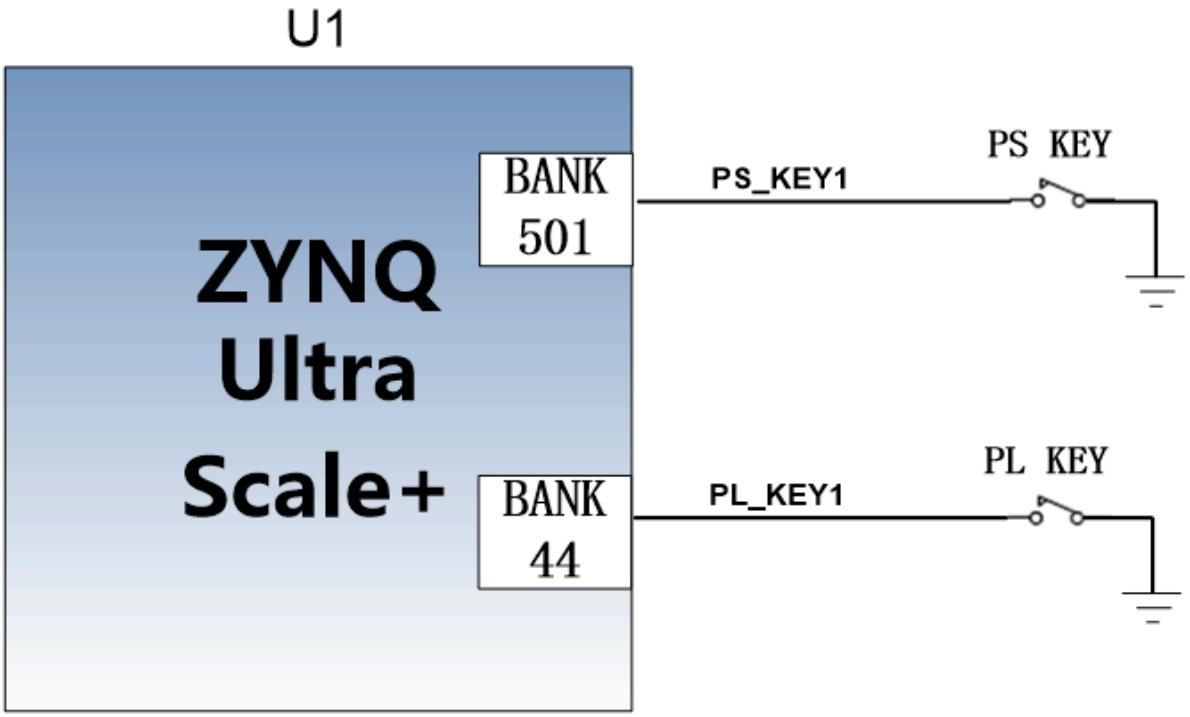


Figure 3-20-1 schematic diagram of the reset button connection

ZYNQ pin assignment for buttons

Signal name	Pin name	Pin number	Remarks
PS_KEY1	PS_MIO26	L15	PS key 1 input
PL_KEY1	B44_L1_N	AE14	PL key 1 input

2.21

DIP switch configuration

There is a 4-bit DIP switch SW1 on the development board to configure the startup mode of the ZYNQ system. The AXU4EV-P system development platform supports 4 startup modes. These 4 startup modes are JTAG debugging mode, QSPI FLASH, EMMC and SD2.0 card startup mode. After the ZU4EV chip is powered on, it will detect (PS_MODE0~ 3) levels to determine which startup mode. Users can choose different startup modes through the DIP switch SW1 on the expansion board. The SW1 startup mode configuration is shown in the table 3-17-1 below.

SW1	Dial position (1, 2, 3, 4)	MODE[3:0]	Startup mode

	ON, ON, ON, ON	0000	PS JTAG
	ON, ON, OFF ,ON	0010	QSPI FLASH
	ON, OFF, ON, OFF	0101	SD card
	ON, OFF, OFF, ON	0110	EMMC

Table 3-21-1 SW1 Startup Mode Configuration

2.22

Power supply

The power input voltage of AXU4EV-P development board is DC12V. On the baseplate, it is converted to + 5V, + 3.3V, + 1.8V through 1 DC/DC power chip TPS54620 and 2 DC/DC power chips MP1482. In addition, the baseplate generates + 1.2V through LDO to power the core board BANK65 (the power supply of BANK65 is selected through the jumper cap. When debugging the MIPI camera, 1.2V power supply is selected, and when using the FMC connector, 1.8V power supply is used). The power supply of BANK66 is + 1.8V. The power supply design diagram on the board is shown in the following figure 3-18-1:

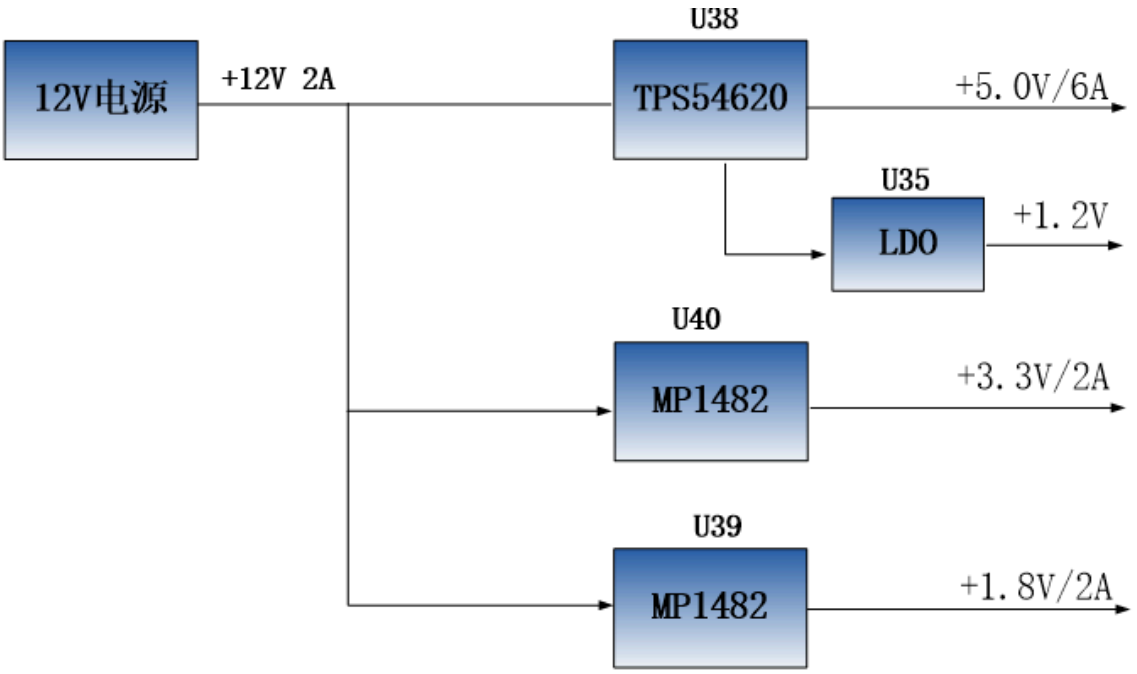


Figure 3-22-1 Power interface part in schematic

The functions of each power distribution are shown in the following table:

Power supply	Function

+5.0V	USB power supply
+1.8V	Ethernet, USB 2.0, core board BANK66
+3.3V	Ethernet, USB 2.0, SD, DP, CAN, RS485
+1.2V	Core board BANK65

2.23

Fan

Because ZU4EV generates a lot of heat when it works normally, we have added a heat sink and fan to the chip on the board to prevent overheating. The control of the fan is controlled by the ZYNQ chip. The control pin is connected to the IO of BANK44 (AG14). If the IO level output is high, the MOSFET tube is turned on and the fan works. If the IO level output is low, the fan stops. The fan design diagram on the board is shown in the following figure 3-23-1:

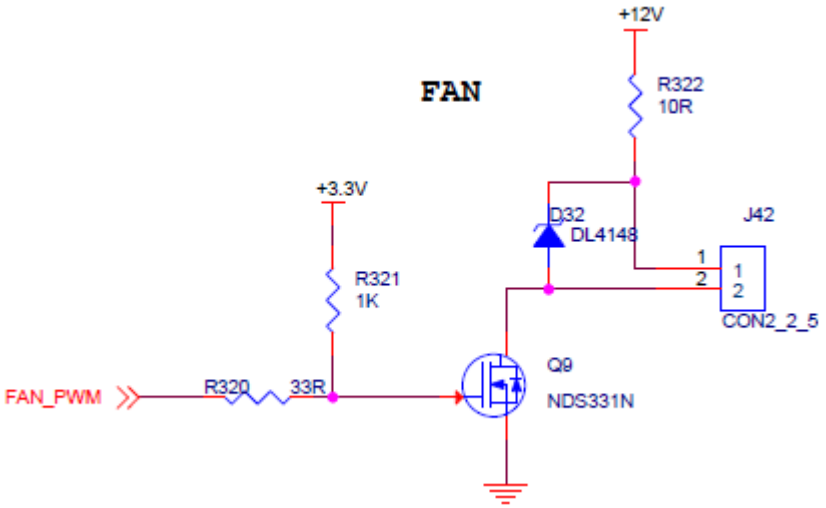
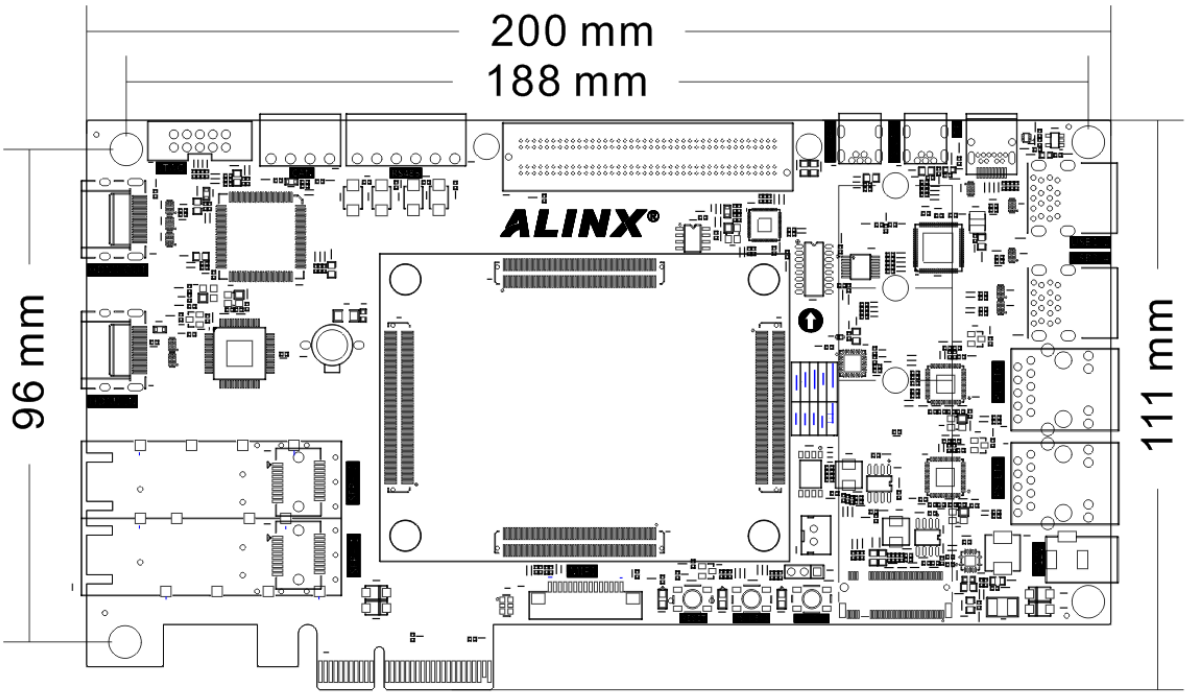


Figure 3-23-1 Fan design in the schematic diagram of the development board

The fan has been fixed to the development board with bolts before leaving the factory. The power supply of the fan is connected to the socket of J42. The red one is the positive pole and the black one is the negative pole.

2.24

Structural dimension diagram



6.

The timing constraint wizard analyzes the clock in the design. Here, the "sys_clk_p" frequency is set to 200Mhz, and then click "Skip to Finish" to end the timing constraint wizard.

7.

Click "OK" in the pop-up window.

8.

Click "Finish"

8.

The eye diagram is configured by default. Note: There may be differences in measuring the eye diagram when using different software versions.

4.

Experimental summary

IBERT provides powerful debugging functions. This experiment only demonstrates some of the functions. For more detailed usage, please refer to the xilinx document pg173.